TRAINER SLIDES · IBF-STS ACCREDITED

Machine Learning 101 for Financial Trading

FSE-DAT-3019-1.1 · Data Analytics and Computational Modelling · Proficiency Level 3

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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Attendance and IBF-STS Funding

1

Sign in every session

Mark attendance at the start of each session and after lunch.

%

75% attendance

IBF-STS funding requires at least 75% attendance.

✓

Pass the assessment

Funding also requires passing both assessment components.

!

Assessment attendance

A separate record is signed before the assessment begins.

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About the Trainer



Complete this panel with your own details before delivery.

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Course Developer · Trainer · Assessor
for this IBF-STS programme.

ROLE
Course Developer, Trainer and Assessor

CERTIFICATION
ACTA · DACE · PhD

FINANCIAL SERVICES
IBF-STS programmes in AI, algorithmic trading and fintech

DELIVERS
AI, machine learning and data programmes for financial institutions

FOUNDER
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ICE-BREAKER

Let's Know Each Other

- Your name, your institution and your role.
- Do you trade, invest, analyse or build systems for markets today?
- Have you used TradingView, a broker API or Python before?
- One trading rule you believe works — we will test it this week.

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HOUSEKEEPING

Ground Rules

Be on time
Sessions restart promptly after each break.

Phones on silent
Take calls outside the room.

Ask as we go
Questions are welcome at any point.

Hands on the keyboard
Twenty labs — follow along on your own laptop.

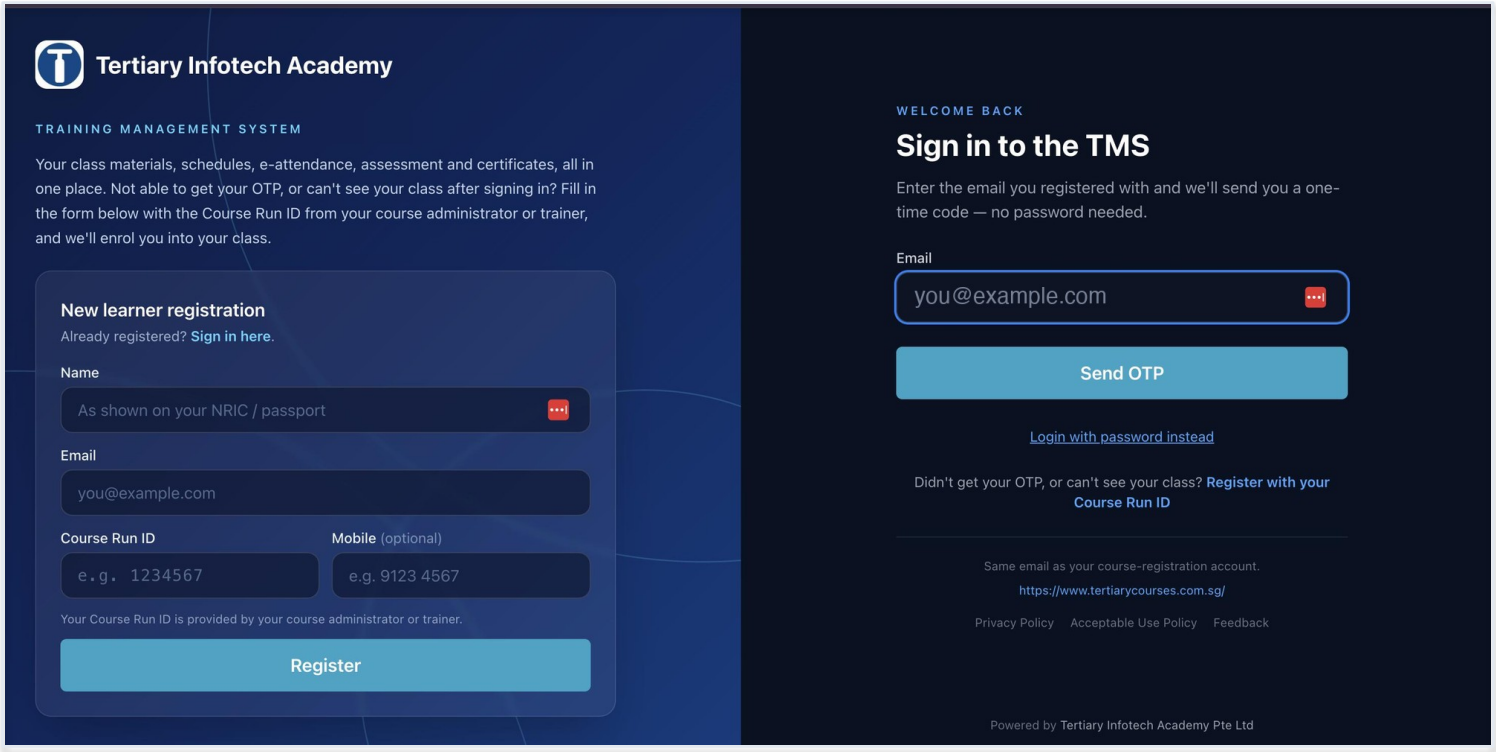
Paper trading only
No real money is used anywhere in this course.

Not investment advice
Lion City Capital and all scenarios are fictitious.

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COURSE PORTAL

Login to TMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email + the OTP from your inbox, then download the slides, Learner Guide and labs.

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COURSE PORTAL · STEPS

Download Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Enter your registered email and click Send OTP
3. Key in the one-time password from your inbox
4. Open this course: download the Learner Slides + Learner Guide PDFs
5. Activities / Lab: download the labs folder (lab manuals + code)

Keep the Learner Guide open all course — the assessment is open book.

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COURSE PORTAL · GITHUB

Download the Labs and Simulator (GitHub)

Course repository

github.com/tertiarycourses/TGS-2023018794-Machine-Learning-101-for-Financial-Trading

Learner Guide, Learner Slides, Lesson Plan and all 20 lab folders (instruction PDFs, .py and .pine files).

Code > Download ZIP, or git clone.

Trading simulator software

github.com/alfredang/algorithmic-trader-bot

The quant / algo trading simulator the labs run in: backtest, forward test, paper trading, analytics.

Install steps: next section and the Learner Guide.

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LEARNING OUTCOMES

What You'll Achieve

1

LO1 · A1, A6

Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.

2

LO2 · A2, A3

Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.

3

LO3 · A4, A5

Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.

4

LO4 · A7, A8, A9, A10

Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.

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COURSE JOURNEY

The Course Journey

1

Topic 1 — Quant and Algo Trading Foundations

Day 1 · Labs 1–4 · A1, A6, K1, K9 · Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents

2

Topic 2 — Indicators, Strategies and Metrics

Day 1 · Labs 5–9 · A2, A3, K3, K5, K7 · Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca

3

Topic 3 — ML Models and A/B Testing

Day 2 · Labs 10–14 · A4, A5, K2, K4, K6 · Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals

4

Topic 4 — End-to-End Trading Pipeline

Day 2 · Labs 15–20 · A7, A8, A9, A10, K7, K8 · Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis

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AI writes the code. You own the algorithm.

In this course the code is written by AI - AI coding assistants and the simulator software do the programming. What matters is the big concepts: the hypothesis, how each algorithm makes its decision, how it is tested without bias and with costs, how risk is controlled, and how the results are judged.

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THE PARADIGM SHIFT

Your Role versus the AI's Role

1

Your job: the hypothesis

What edge do you expect, why, and how will you know it is real?

2

Your job: the algorithm's logic

The rule or model, its inputs, and when it buys or sells.

3

Your job: testing and judgement

Out-of-sample, next-bar, net of costs; read the metrics.

4

Your job: risk and accountability

Limits, kill switch, approval - you answer for every trade.

AI

AI's job: the code

AI assistants and the simulator write, run and fix the Python and Pine.

AI

AI's job: the drudgery

Data plumbing, charts, reports - you review what it produces.

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TSC Alignment — Abilities

1 **A1**
Develop hypotheses for implementing data analytics and computational modelling activities

2 **A2**
Identify appropriate statistical algorithms and data models to test hypotheses and/or theories

3 **A3**
Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements

4 **A4**
Conduct statistical modelling of data to derive patterns and solutions

5 **A5**
Perform A/B testing based on different hypotheses

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TSC Alignment — Abilities (continued)

1. **A6**
   Identify possible big data applications

2. **A7**
   Perform coding and configuration of software agents or programs based on selected models or algorithms

3. **A8**
   Conduct tests on the actions taken and outcomes to assess effectiveness of the models

4. **A9**
   Diagnose unintended outcomes produced by statistical models

5. **A10**
   Draw relevant trends and insights from data analysis to support decisions and report accordingly

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TSC Alignment — Knowledge

1. **K1**
   Data analytics and modelling business use cases
2. **K2**
   Types of statistical analyses, data models, algorithms and advanced computational methods
3. **K3**
   Range and application of various statistical methods and algorithms
4. **K4**
   Range and application of various types of data models
5. **K5**
   Usage of analytics platforms and tools

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TSC Alignment — Knowledge (continued)

1. **K6**
   Statistical modelling techniques

2. **K7**
   Coding languages for programming of algorithms and signals

3. **K8**
   Potential reasons for unintended outcomes

4. **K9**
   Forms of data within financial services

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COURSE SCHEDULE

Lesson Plan — Day 1 (1 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

09:00 - 09:30	Course Overview, Trainer Introduction and Ground Rules	
09:30 - 10:45	T1.1-1.4 Quant Success Stories, the Evolution of Quant, the ML-to-LLM Shift, and Agentic AI versus AI Agents	K1, K9
10:45 - 11:00	**Tea Break**	
11:00 - 11:30	T1.5-1.6 Forms of Financial Data, Big Data Applications and Trading Hypotheses	A1, A6, K1, K9
11:30 - 11:50	**Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account**	K5
11:50 - 12:10	**Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView**	A3, K5, K9
12:10 - 12:30	**Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality**	A6, K9
12:30 - 13:30	**Lunch Break**	
13:30 - 13:45	**Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test**	A1, A6, K1
13:45 - 14:30	T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies	A2, K3

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COURSE SCHEDULE

Lesson Plan — Day 1 (2 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

14:30 - 14:55	**Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins**	A3, K3, K7
14:55 - 15:25	**Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation**	A2, A3, K3, K7
15:25 - 15:40	**Tea Break**	
15:40 - 16:05	**Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals**	A2, K3, K7
16:05 - 16:50	T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement	A3, K5, K7
16:50 - 17:20	**Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session**	A3, K5
17:20 - 17:50	T2.5 Performance and Risk Metric Definitions	A3, K3
17:50 - 18:15	**Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table**	A3, K3
18:15 - 18:30	Day 1 Recap and Q&A	

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COURSE SCHEDULE

Lesson Plan — Day 2 (1 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

09:00 - 09:15	Day 1 Recap and Day 2 Overview	
09:15 - 09:40	T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models	A4, K2, K4
09:40 - 10:00	**Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay**	A2, A4, K2
10:00 - 10:20	**Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position**	A4, K4, K6
10:20 - 10:35	**Tea Break**	
10:35 - 10:55	T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques	A4, K4, K6
10:55 - 11:10	**Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals**	A4, K4, K6
11:10 - 11:25	T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm	A4, A5, K2, K6
11:25 - 11:40	**Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves**	A5, K2
11:40 - 12:00	**Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve**	A4, K6
12:00 - 12:30	T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs	A7, A8, K7
12:30 - 13:30	**Lunch Break**	

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COURSE SCHEDULE

Lesson Plan — Day 2 (2 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

13:30 - 13:55	**Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even**	A7, A8, K7
13:55 - 14:05	T4.3 Forward Testing with Automated Paper Trading	A7, A8
14:05 - 14:30	**Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading**	A7, A8, K7
14:30 - 14:50	**Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution**	A7, K5, K7
14:50 - 15:00	T4.4 Potential Reasons for Unintended Outcomes	A9, K8
15:00 - 15:20	**Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live**	A9, K8
15:20 - 15:35	**Tea Break**	
15:35 - 15:45	T4.5 Performance Analysis, Insights and Reporting	A10
15:45 - 16:10	**Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report**	A8, A10
16:10 - 16:30	**Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval**	A7, A10, K1
16:30 - 17:15	**Written Assessment (Short Answer Questions)**	K1 - K9
17:15 - 18:30	**Practical Performance Assessment**	A1 - A10

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BEFORE THE ASSESSMENT

Briefing for Assessment

Open book
You may use your Learner Guide, lab code and your own notes.

Two components
Written Assessment (45 min) and Practical Performance (75 min).

Both must be passed
Competent in BOTH is required to be awarded the module.

All your own work
Collaboration results in a Not Yet Competent outcome.

Ask if unclear
The assessor can explain a question, but not how to answer it.

Not Yet Competent?
You will be counselled on the gaps and offered a re-assessment.

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Final Assessment

Written Assessment

Short Answer Questions · 45 minutes · open book.

9 open-ended questions — one for each knowledge statement K1 - K9.

No multiple-choice questions.

Practical Performance

Python + Alpaca paper trading · 75 minutes · open book.

4 practical tasks covering ability statements A1 - A10.

A new trading scenario, using the same pipeline tool-kit as the labs.

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ON ASSESSMENT DAY

Assessment Flow

1. Sign the assessment attendance record
2. Assessor briefing (not assessed)
3. Sit the WA (45 min) then the PP (75 min) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

Both components are open book. You must be Competent in BOTH to be awarded the module.

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GETTING STARTED

01

Set Up Alpaca and TradingView

LEARNING OUTCOME

Create the free accounts and the Python environment every lab uses. Everything runs on paper — no real money at any point.

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GETTING STARTED

Download, Install and Start the Trading Simulator Software

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Install Git, Python 3.11+ and uv (or Docker)
2. git clone the simulator - or Download ZIP
3. Copy .env.example to .env
4. uv sync > uv run trader seed > uv run trader serve --with-worker
5. Or: docker compose up -d --build > docker compose exec backend trader seed
6. Open http://localhost:8000 > Course Labs

Simulator: github.com/alfredang/algorithmic-trader-bot - the README has the full start-up steps and troubleshooting.

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GETTING STARTED

Sign Up for Alpaca and Open the Paper-Trading Account

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Sign up at alpaca.markets and verify your email
2. Log in; choose 'Paper Trading' (upper left)
3. API section: generate key - copy the secret once
4. Paste keys into labs/.env
5. Run python -m pipeline check - must say ONLINE

No funding needed. Paper keys only work on paper-api.alpaca.markets. No account? Every lab also runs offline.

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GETTING STARTED

Sign Up for TradingView and Find the Tools You Will Use

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Sign up at tradingview.com (email, Google or Apple)
2. Keep the free plan
3. Products > Supercharts; search NASDAQ:AAPL or SGX:D05
4. Set time zone to Singapore
5. Find Indicators, Pine Editor, Strategy Tester, Trade

The free plan covers Labs 2, 5 and 6. Webhook alerts in Lab 17 need a paid plan - the lab includes a local simulator.

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PLATFORM · THE IDEA THAT MATTERS MOST

Paper first. Always.

Every strategy in this course is tested on history, then forward-tested on a paper account with automatic orders, before anyone would consider real capital. That discipline is the course.

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TOPIC 1

01

Quant and Algo Trading Foundations

LEARNING OUTCOME

Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.

Abilities A1, A6 · Knowledge K1, K9 · Day 1 · Labs 1–4 · Assessed by PP Task 1 and WA Q1, Q9

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Quant Success Stories

1

Ed Thorp - Princeton Newport · 1969

First market-neutral quant fund: about 20% a year after fees for two decades by hedging warrants and convertibles

2

Renaissance - Medallion · 1988

Purely model-driven; about 66% a year before fees, 39% after (1988-2018); closed to outsiders, employee-owned

3

D.E. Shaw · 1988

Founded by a computer-science professor from Morgan Stanley's quant group; models and software; about US$60-65bn

4

Man AHL · 1987

Systematic trend following from hand-typed price histories to momentum models across 800+ markets

5

Two Sigma · 2001

AI, machine learning and distributed computing applied to trading; about US$70bn under management

6

Jane Street · 2024

Quant market maker: about US$20.5bn net trading revenue; nearly a quarter of primary US ETF activity

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Spotlight: Renaissance Technologies and the Medallion Fund

1

Founded by a mathematician

Jim Simons founds Monemetrics in 1978; renamed Renaissance Technologies in 1982

2

Medallion fund, 1988

The flagship fund trades purely on mathematical models of market data

3

About 66% a year before fees

Average gross return 1988-2018, about 39% after its high fees; closed to outsiders, employee-owned

4

The lesson for this course

The edge came from data, rigorous testing, costs and risk control - not from one clever indicator

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What the Success Stories Have in Common

1. **Hypothesis first**
   Each started from a testable idea about prices, not from a tool

2. **Data and testing**
   Clean data and out-of-sample tests before capital

3. **Many small edges**
   Diversified, market-neutral bets rather than one big call

4. **Costs and risk**
   Execution cost, leverage and risk limits decide survival - see LTCM 1998

5. **Citadel Securities**
   Quant market maker estimated to handle about a quarter of all US equities trading

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TOPIC 1 · DEFINITION

Algorithmic trading

Using computer programs that follow predefined rules on price, timing, volume or models to generate and execute orders automatically, with limited or no human intervention.

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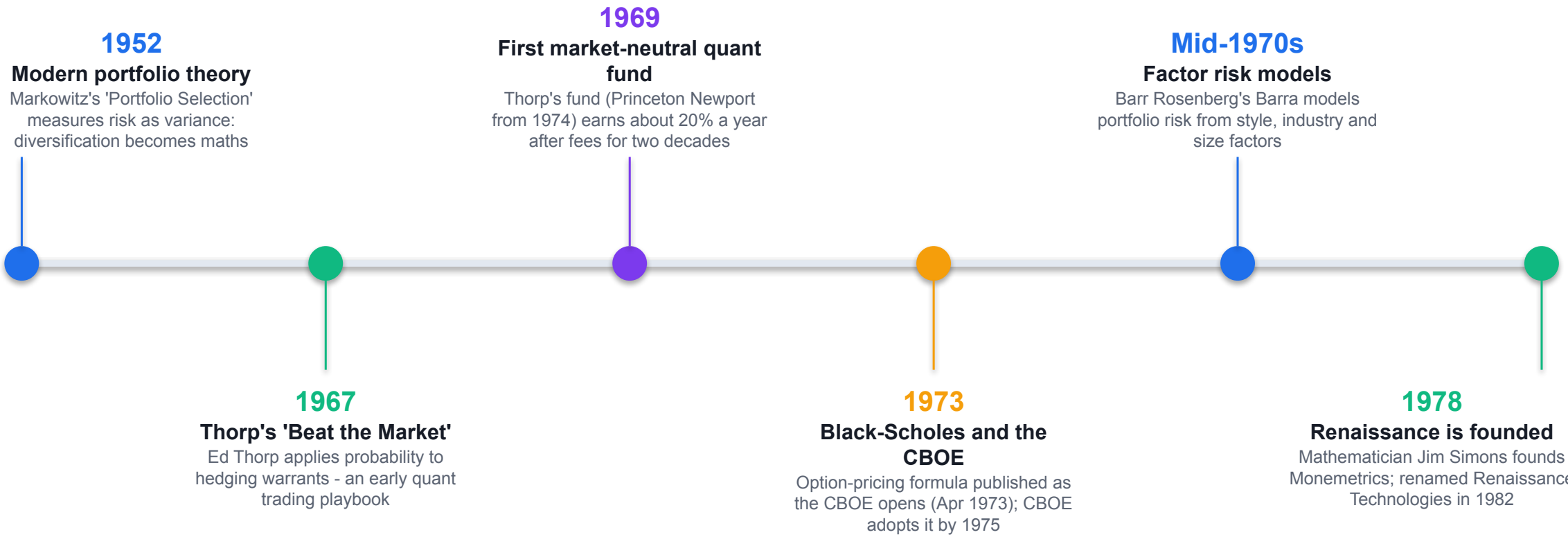
The Evolution of Quant: From Thorp to HFT to AI

1. 1950s-70s: portfolio theory, arbitrage and option pricing
2. 1970s-80s: factor risk models and program trading
3. 1980s-90s: statistical arbitrage and pairs trading
4. 1988 on: fully systematic signal mining (Renaissance, D.E. Shaw)
5. 2000s: electronic market making and HFT
6. 2010s: speed race - co-location, microwave, microseconds
7. 2020s: ML on alternative data, then LLM and agentic quant

Each era kept the last era's discipline - hypothesis, data, testing, costs, risk - and added speed or new data.

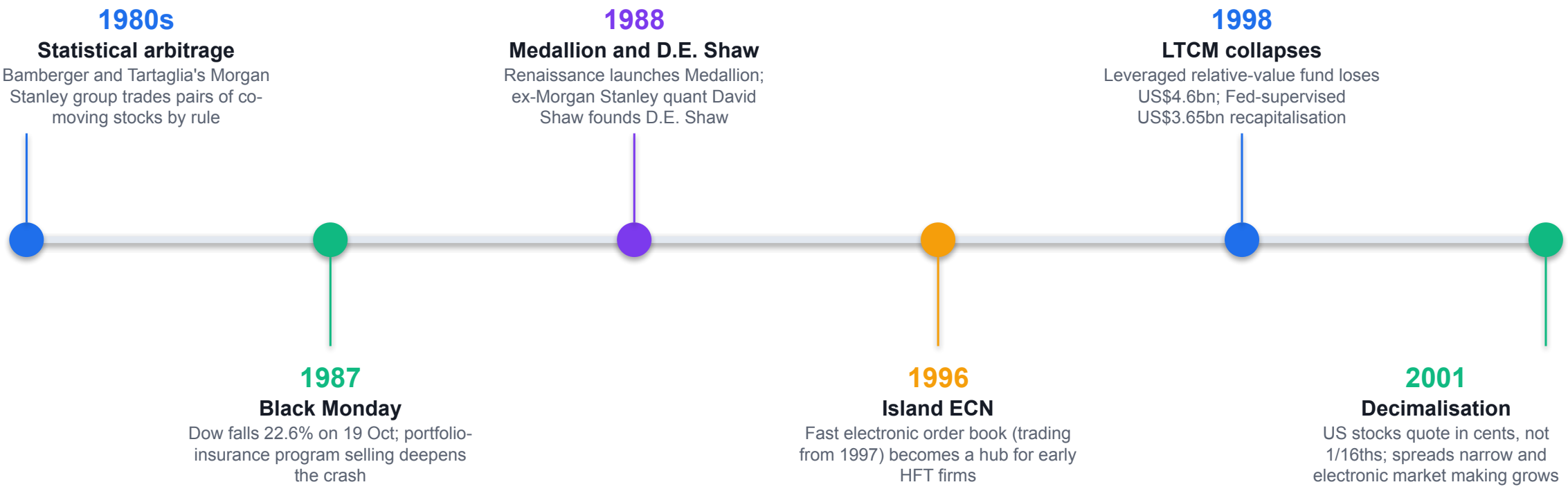
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The Evolution of Quant and Algorithmic Trading (1 of 4)



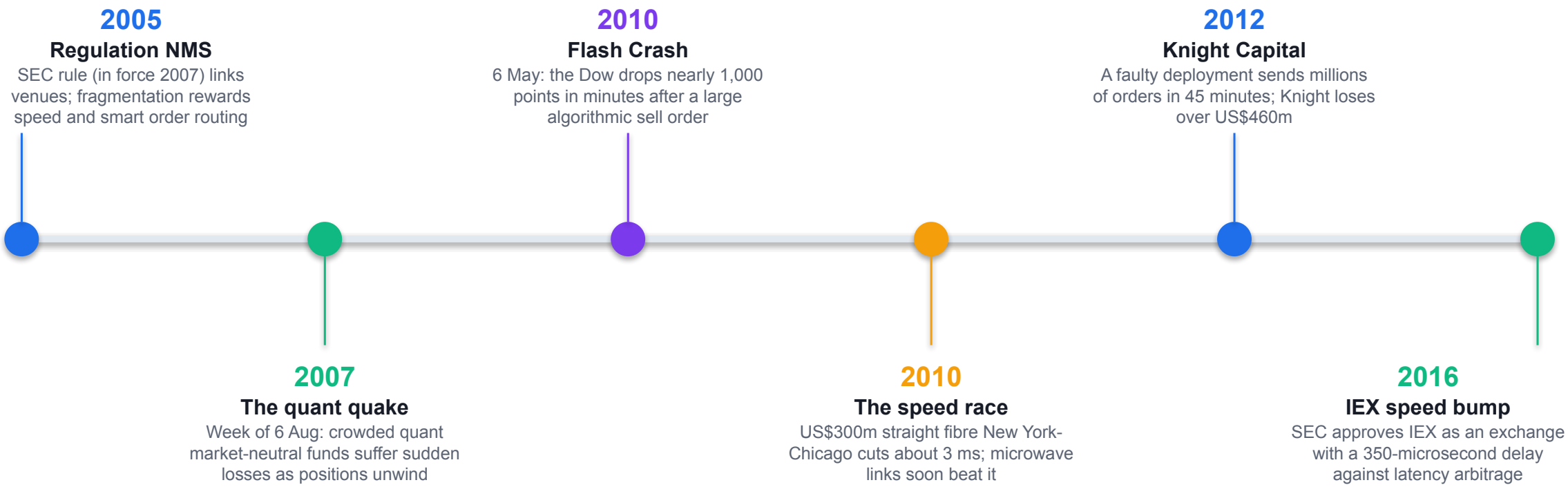
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The Evolution of Quant and Algorithmic Trading (2 of 4)



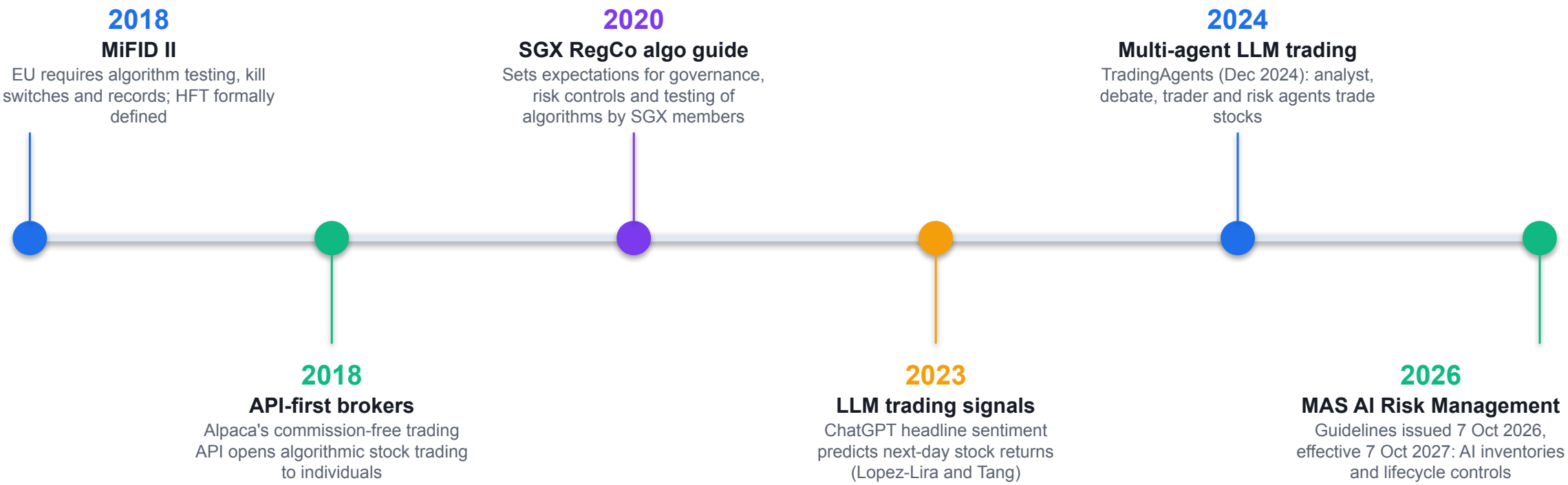
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The Evolution of Quant and Algorithmic Trading (3 of 4)



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The Evolution of Quant and Algorithmic Trading (4 of 4)



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Six Paradigm Shifts in Quant and Algorithmic Trading

1

From intuition to probability · 1950s-1970s

Markowitz, Thorp and Black-Scholes make risk, hedging and option value mathematical

2

From stocks to factors · 1970s-1980s

Barra factor models and program trading treat portfolios as baskets of risk exposures

3

From factors to statistical arbitrage · 1980s-2000s

Morgan Stanley's pairs group, D.E. Shaw and Renaissance harvest many small market-neutral edges

4

From floors to electronic speed · 1996-2010s

ECNs, decimalisation and Reg NMS enable HFT: co-location, microwave links, microseconds

5

From statistics to machine learning · 2000s-2010s

Models learn non-linear patterns from prices, order books and alternative data

6

From writing code to directing AI · 2023-present

AI writes and runs the code; quants own the big concepts - hypothesis, algorithm logic, testing, risk

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Why Algorithms - and What Can Go Wrong

Benefits

- Speed: reacts and executes in milliseconds
- Discipline: rules remove fear, greed and fatigue
- Backtesting: test ideas on historical data first
- Lower costs: split large orders to reduce market impact
- Scale: monitors many markets and assets around the clock

Risks

- Technical failure: bugs and outages (Knight Capital, 2012)
- Overfitting: great backtests that fail on live data
- Black swans and regime change the model has never seen
- Feedback loops between algorithms (2010 Flash Crash)
- Set-up, data and compliance costs; no guaranteed profit

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The Eras of AI in Trading

1

Symbolic AI · 1956-1980s

Hand-written rules: expert systems and rule-based program trading; brittle when markets change

2

Machine learning · 1990s-2000s

Models learn from data: regression, tree and boosting signals; statistical arbitrage at scale

3

Deep learning · 2012-2017

Many-layer networks on GPUs find non-linear patterns in prices, order books and alternative data

4

Generative AI · 2017-2023

Transformers and LLMs turn news, filings and earnings calls into trading signals

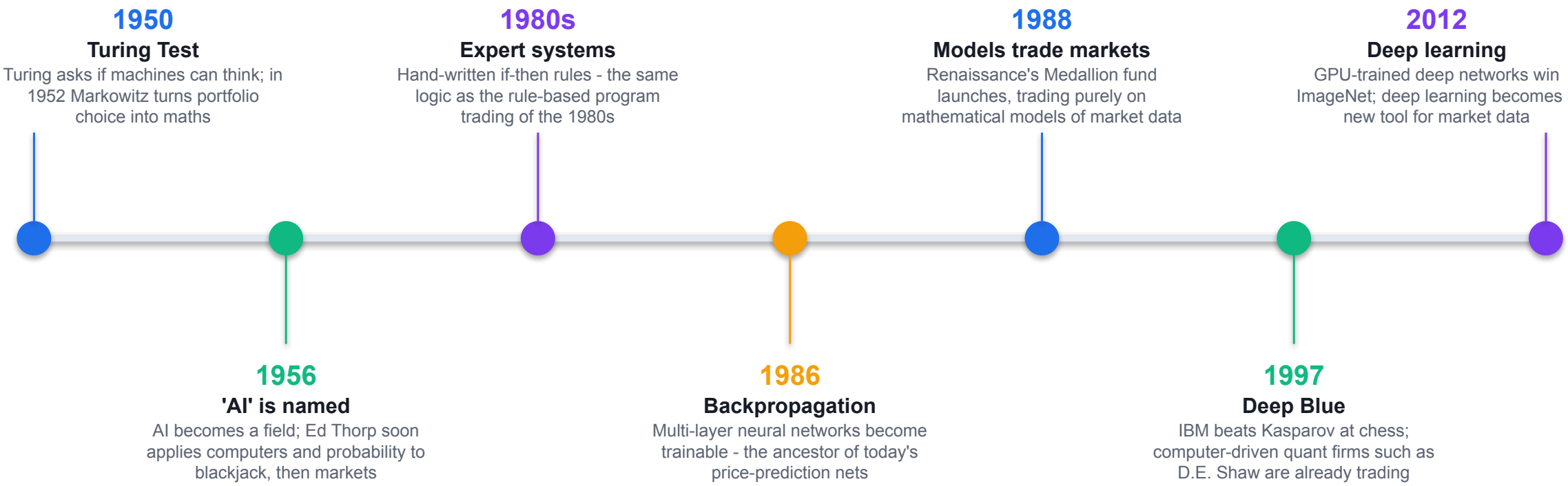
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Agentic AI · 2024-present

LLM agents research, code, backtest and paper-trade strategies under hard risk limits

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AI Milestones and Their Trading Counterparts (1 of 2)



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AI Milestones and Their Trading Counterparts (2 of 2)



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Paradigm Shift: Machine-Learning Quant versus LLM-Era Quant

	Machine-learning quant	LLM-era quant
Main data	Structured: prices, volumes, fundamentals	Adds unstructured text: news, filings, earnings calls, research
Features	Hand-engineered by the quant (lags, RSI, factors)	Text read directly; LLM scores sentiment and events
Model building	Fit regressions, trees, boosting on labelled data	Prompt or fine-tune pretrained LLMs; agents propose alphas
Who writes the code	The quant, line by line	AI writes it; the quant directs and reviews
Research workflow	The quant writes every line of code	Coding agents draft code and backtests; the quant reviews
Alpha decay	Signals fade as more funds trade them	LLM signals fade too as LLM adoption rises
New risks	Overfitting; look-ahead in features	LLM training-data look-ahead and 'distraction'; homogeneous factors
Human role	Model builder	Supervisor: hypothesis, testing, risk sign-off

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Impact of LLMs on Quant

1

Text becomes a signal

ChatGPT headline scores predict next-day returns, strongest for small stocks and bad news (Lopez-Lira and Tang)

2

Finance-specific LLMs

BloombergGPT (2023): 50bn parameters trained with 363bn tokens of Bloomberg financial data

3

Automated alpha mining

Alpha-GPT turns ideas into formulaic alphas; AlphaAgent targets alpha decay and crowding

4

Backtests need new controls

LLMs may 'know' in-sample history: test after the knowledge cutoff and anonymise firms (Glasserman and Lin)

5

Faster crowding

Strategy returns decline as LLM adoption rises - prices become more efficient (Lopez-Lira and Tang)

6

Agents still need limits

Agents propose trades; hard-coded risk checks and humans approve them

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Agentic AI versus AI Agents in Trading

Agentic AI - the capability

The capability of AI to plan and act: it calls tools, uses skills and memory, and follows instructions to complete multi-step work when it is asked.

Trading example: On request, an LLM fetches a stock's bars and news through tools, checks its memory of past trades and proposes an order within its instructions

AI Agent - always on, 24/7

A special class of agentic AI that stands by 24/7 and works all the time: always on, woken by schedules, events and messages, with its own long-term memory.

Examples: OpenClaw, Hermes Agent, OpenAI Dots.

Trading example: A round-the-clock trading agent that watches prices and news overnight and proposes paper trades within risk limits

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AI Agents vs Agentic AI - Side by Side

	AI Agents	Agentic AI
What it is	A special class of agentic AI that is always on	Capability to plan and act with tools, skills, memory
Runtime	Persistent: stands by 24/7 on a server or PC	Runs when invoked: a chat, workflow or pipeline
Trigger	Schedules, events, messages - unprompted	A user prompt or a pipeline step
Memory	Long-term memory across days and tasks	Session or task memory; can use long-term memory
Building blocks	Agentic AI + always-on runtime and scheduler	LLM + tools + skills + memory + instructions
Examples	OpenClaw, Hermes Agent, OpenAI Dots	Claude Code, ChatGPT with tools, TradingAgents
Trading example	24/7 market and news monitor proposing trades	On-demand multi-agent analysis before a trade
Main risks	Acting unsupervised overnight; runaway loops	Hallucination; prompt injection; tool misuse

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From Chatbot to Always-On AI Agent

1

1 · Chatbot: Answers one prompt at a time; no tools, no actions

2

2 · Agentic AI: Plans and calls tools, skills, memory and instructions when asked

3

3 · Multi-agent agentic system: Specialist agents coordinate: analysts, debate, trader, risk manager

4

4 · AI agent (always on): Stands by 24/7, wakes on schedules and events, works continuously within limits

An AI agent is agentic AI that never switches off - so its limits, kill switch and monitoring matter most.

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Roles in a Multi-Agent Trading System

1. **Fundamental Analyst**
   Reads filings, earnings and balance sheets to judge intrinsic value

2. **Technical Analyst**
   Reads price, volume and indicators such as MACD and RSI

3. **Sentiment Analyst**
   Gauges mood from social media, analyst ratings and options flow

4. **News Analyst**
   Tracks news wires, press releases and macro events

5. **Bull / Bear Researchers**
   Argue for and against the trade so weak ideas are challenged

6. **Trader**
   Combines the reports into a buy, sell or hold call with timing and size

7. **Risk Manager**
   Checks volatility, liquidity, exposure and drawdown limits

8. **Portfolio Manager**
   Approves or rejects the proposal and fits it into the whole portfolio

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Stock-Trading Agent Frameworks (1 of 2)

1

TradingAgents (Xiao et al., 2024)

Simulated stock-trading firm: fundamental, sentiment, news and technical analysts, bull/bear debate, trader, risk team

2

FinMem (Yu et al., 2023)

LLM trading agent with profiling and layered memory making Buy / Sell / Hold decisions on individual stocks

3

FinAgent (Zhang et al., 2024)

Multimodal agent reading prices, news and chart images, with dual-level reflection; tested on stocks and crypto

4

StockAgent (2024)

LLM investor agents trade a simulated stock market under macro, policy and event shocks; avoids test-set leakage

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Stock-Trading Agent Frameworks (2 of 2)

1

QuantAgent (2025)

Price-driven multi-agent system for high-frequency trading: Indicator, Pattern, Trend and Risk agents

2

AI Hedge Fund (open source)

Investor-persona agents (e.g. Warren Buffett), a risk manager and a portfolio manager decide trades for chosen tickers

3

Agentic Trading Lab (Open-Finance-Lab)

Agent-run stock portfolios with backtests and paper trading through the Alpaca API

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Risks and Controls for AI Agents in Trading

	What goes wrong	Control
Hallucination	An LLM states a wrong earnings figure or invents a signal with confidence	Ground answers in tool or RAG data, log the reasoning, review unusual calls
Prompt injection	Malicious text in news, web pages or files hijacks the agent's instructions	Input and output guardrails, least-privilege API keys, no secrets in prompts
Overtrading and runaway loops	Agents trade too often or amplify each other, compounding losses	Order and position caps, daily-loss circuit breaker, kill switch
Model and backtest risk	Overfitting, look-ahead bias and regime change: strong backtests, weak live results	Walk-forward tests, paper trade first, monitor drift and costs
Accountability	Unclear who answers for an autonomous agent's trade or a vendor's AI	The firm stays accountable: human sign-off, audit trails, clear owners
Regulation (MAS)	FEAT principles (2018) and MAS AI Risk Management Guidelines (issued Oct 2026)	Board oversight, AI inventory, materiality assessment and lifecycle controls

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Forms of Data within Financial Services

1

Market data

OHLCV bars, quotes, trades, order-book depth — Alpaca serves bars, quotes, trades, news.

2

Reference & corporate actions

Symbols, calendars, splits, dividends — unadjusted splits fake a crash.

3

Fundamental & macro

Statements, ratios, earnings; rates, inflation, central-bank decisions.

4

Alternative data

News and social sentiment, web traffic, satellite, card spend, ESG.

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Big Data in Trading — the Four Vs

1

Volume

Tick data runs to billions of rows per year for one exchange.

2

Velocity

Streaming quotes and news arrive in milliseconds.

3

Variety

Prices, text, images, transcripts — structured and unstructured.

4

Veracity

Bad ticks, survivorship, restatements — verify before you trust.

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Big Data Applications

1

Execution analytics

Tick and quote data to measure and cut slippage.

2

Sentiment and event trading

News and social text scored by NLP and LLMs.

3

Regime detection

Cross-asset returns clustered into market states.

4

Surveillance and risk

Order and trade data scanned for market abuse; intraday risk.

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Data Analytics and Modelling Use Cases

1

Signal generation

Find repeatable return — judged by out-of-sample, net Sharpe.

2

Execution

VWAP / TWAP / shortfall — judged by slippage in basis points.

3

Risk forecasting & sizing

Volatility and VaR models — judged by limit breaches.

4

Research at scale

LLMs read filings and news — judged by accuracy and hours saved.

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What Makes a Hypothesis Testable

1

Instrument and universe

e.g. SPY, daily bars, 2018 – 2025.

2

Signal and holding period

e.g. three down closes in a row; hold 5 days.

3

Expected effect and benchmark

Forward return exceeds the unconditional 5-day return.

4

Written BEFORE the test

H0, H1 and the significance level chosen first.

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Machine Learning Implementation Steps

1. Frame the problem and hypothesis
2. Collect and clean the data
3. Define target and features
4. Split by time; train and validate
5. Test out of sample
6. Paper trade, then review

A model that skips the out-of-sample test and the paper trade has not been tested.

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K5

Pipeline Skeleton, Configuration and the Alpaca Paper Account

LAB 01

Get the simulator running, decide between the offline demo and an Alpaca paper account, and check every integration's status. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Load the demo data
2. Simulator: Check the integrations
3. Simulator: Check the AI agent login
4. Simulator: Read the market clock
5. Simulator: Tour the Pipeline Map
6. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

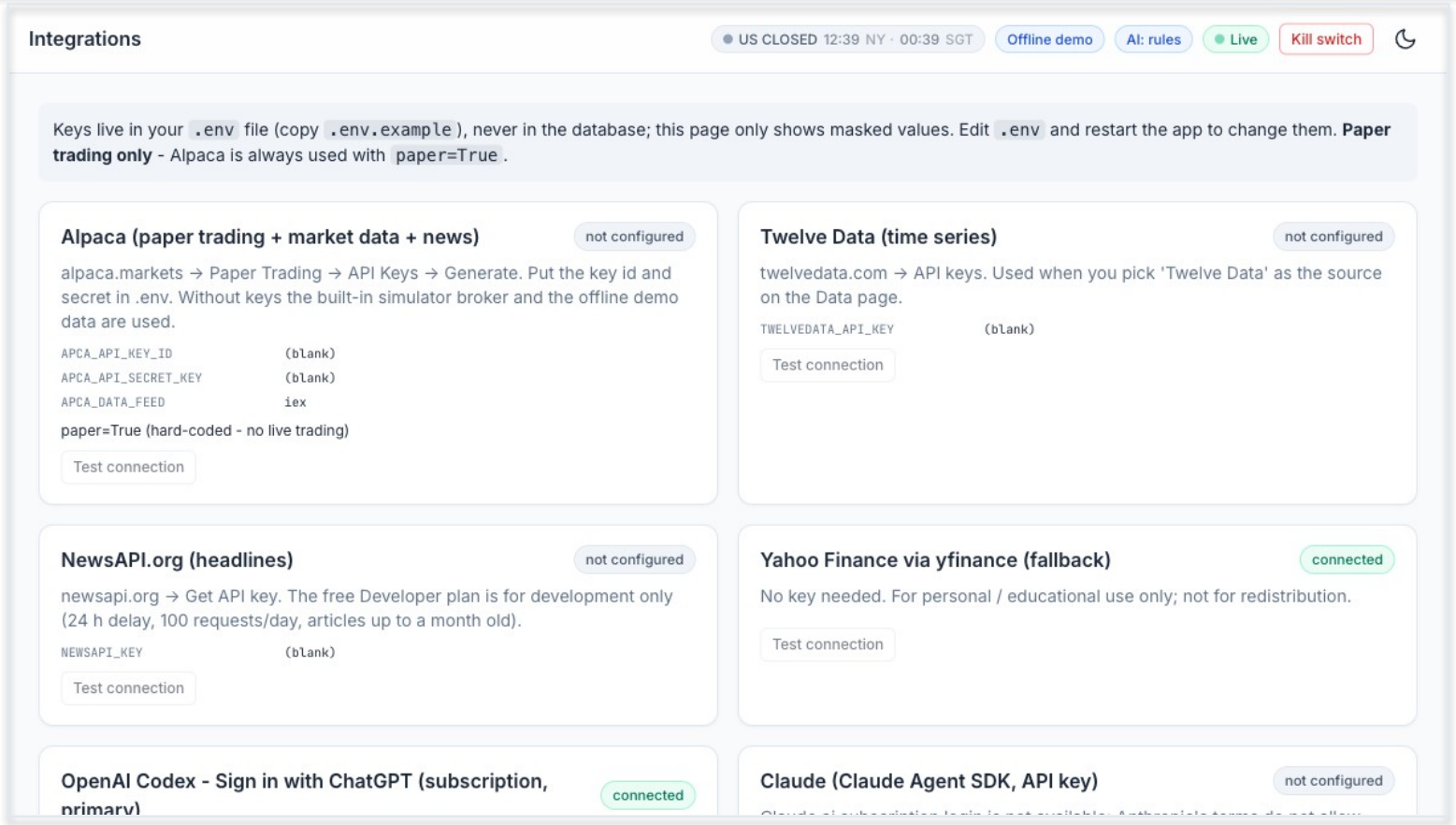
The pipeline skeleton: config.yaml, risk limits and a verified Alpaca paper account (or simulator).

Duration
20 min

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LAB 01 · SIMULATOR

Lab 01 — In the Trading Simulator



Course Labs > Lab 01 at http://localhost:8000 - Settings > Demo data. Offline demo data; connect Alpaca paper keys for live paper trading.

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A3 K5 K9

Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

LAB 02

Prove the app's prices are right: compare its bars with TradingView's Data Window and explain adjusted vs unadjusted prices. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Read TradingView's Data Window
2. Simulator: Compare with RAW and ADJUSTED bars
3. Simulator: Repeat for SGX:D05 (DBS)
4. Simulator: Inspect the chart
5. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

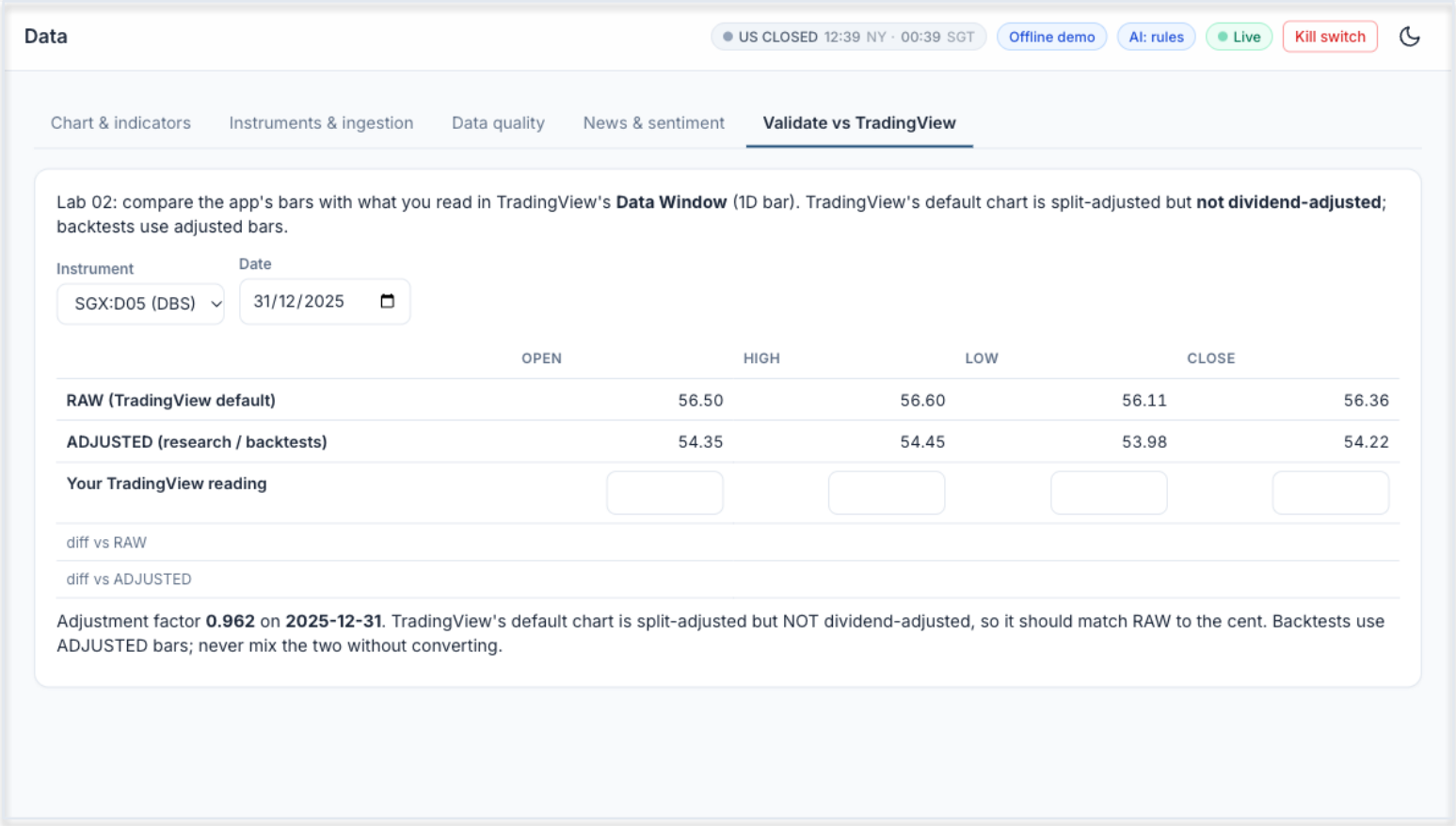
A data-validation report reconciling pipeline bars with TradingView for SGX:D05 and AAPL.

Duration
20 min

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LAB 02 · SIMULATOR

Lab 02 — In the Trading Simulator



Course Labs > Lab 02 at http://localhost:8000 - Data > Validate vs TradingView (AAPL). Offline demo data; connect Alpaca paper keys for live paper trading.

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Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality

LAB 03

Ingest bars from Alpaca / Twelve Data / yfinance (or the demo files), run the data-quality checks and score news sentiment. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Fetch data
2. Simulator: Read the data-quality report
3. Simulator: Switch timeframes
4. Simulator: Headlines and sentiment
5. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

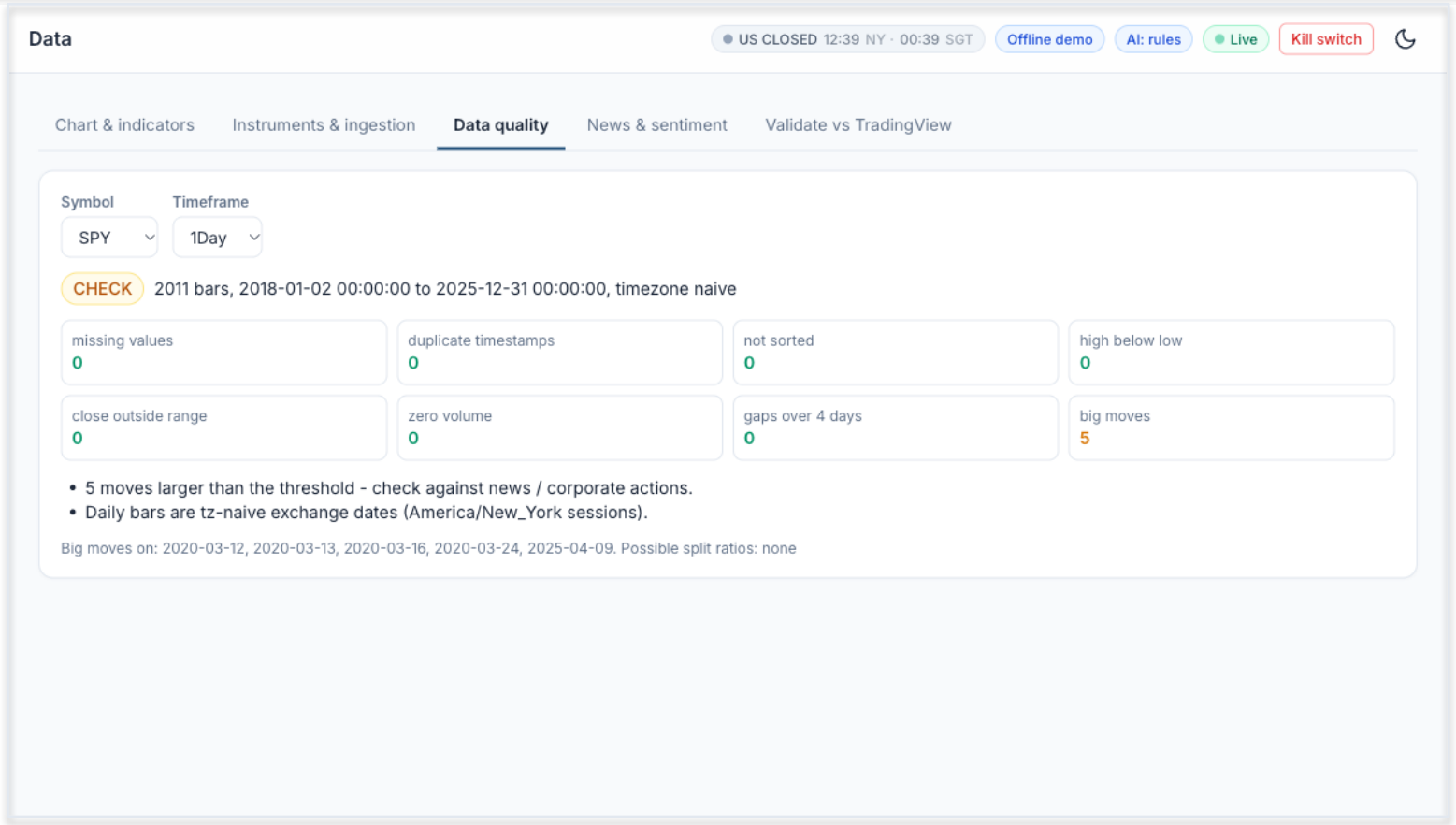
An ingestion stage caching multi-timeframe bars and headlines with automated data-quality checks.

Duration
20 min

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LAB 03 · SIMULATOR

Lab 03 — In the Trading Simulator



Course Labs > Lab 03 at http://localhost:8000 - Data > Fetch AAPL. Offline demo data; connect Alpaca paper keys for live paper trading.

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Lunch Break

1 hour · see you at 13:30

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A1 A6 K1

Hypothesis Stage: from a Written Card to an Automated Statistical Test

LAB 04

Write H0 and H1 before looking at results and test them with a Welch t-test and a bootstrap CI. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Write the hypothesis card
2. Simulator: Run the test
3. Simulator: Robustness and multiple testing
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

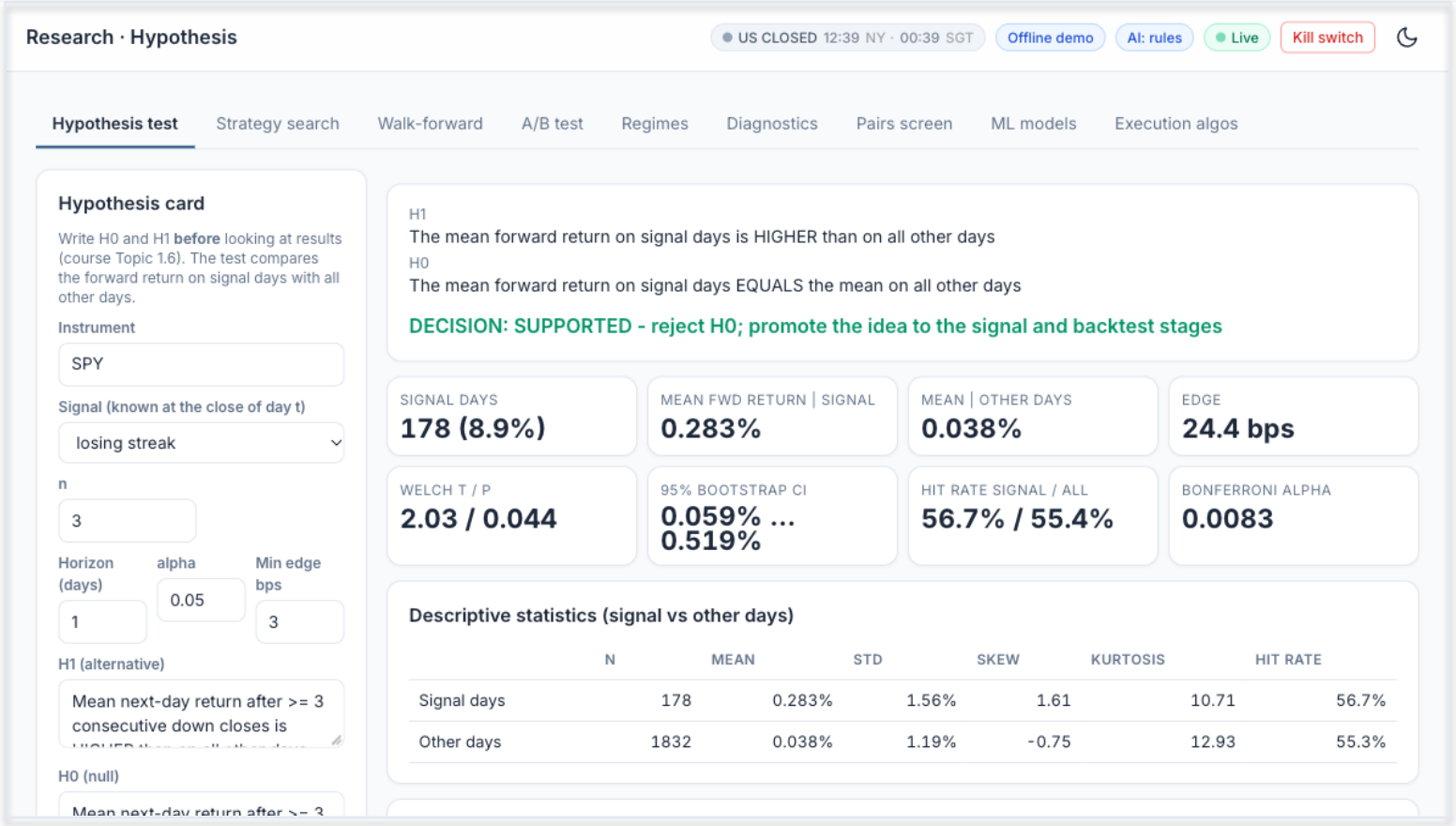
A hypothesis card and an automated statistical test of the SPY dip-after-streak hypothesis.

Duration
15 min

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LAB 04 · SIMULATOR

Lab 04 — In the Trading Simulator



Course Labs > Lab 04 at http://localhost:8000 - Research > Hypothesis (SPY, losing streak 3). Offline demo data; connect Alpaca paper keys for live paper trading.

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A1 A6 K1 K9

Topic 1 Recap

1

A1

Develop hypotheses for implementing data analytics and computational modelling activities

2

A6

Identify possible big data applications

3

K1

Data analytics and modelling business use cases

4

K9

Forms of data within financial services

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TOPIC 2

02

Indicators, Strategies and Metrics

LEARNING OUTCOME

Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.

Abilities A2, A3 · Knowledge K3, K5, K7 · Day 1 · Labs 5–9 · Assessed by PP Task 2 and WA Q3, Q5, Q7

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K3

Trend and Momentum Indicators

1

SMA / EMA

Average close over n bars; EMA weights recent bars by 2/(n+1).

2

MACD

EMA12 − EMA26 with a 9-EMA signal line; crosses flag momentum.

3

RSI (14)

100 − 100/(1+RS); above 70 overbought, below 30 oversold.

4

Donchian channel

Highest high / lowest low of n bars — the breakout level.

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Volatility and Volume Indicators

1

Bollinger Bands

SMA20 ± 2 standard deviations; width tracks volatility.

2

ATR (14)

Average true range — position sizing and volatility stops.

3

VWAP

Volume-weighted average price — the execution benchmark.

4

OBV

Adds volume on up days, subtracts on down days.

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Matching the Method to the Hypothesis

1

'Does the signal beat the benchmark?'

Backtest + t-test or bootstrap on returns.

2

'Will tomorrow be up?'

Classifier: logistic regression, random forest, XGBoost.

3

'What state is the market in?'

Clustering: K-means on return and volatility features.

4

'Which parameters?'

Grid search in-sample, confirm out-of-sample.

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Coding Languages for Algorithms and Signals

1

Pine Script v6

//@version=6 · ta.ema · ta.crossover · strategy.entry.

2

Python

Signals as pandas Series of +1 / 0 / −1 aligned to bars.

3

C++ / Java

Low-latency execution engines and exchange gateways.

4

R, SQL, MQL

Statistics, data stores, MetaTrader expert advisors.

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Order Types and Time in Force

1

Market and limit

Certainty of fill versus control of price.

2

Stop, stop-limit, trailing stop

Trigger on a price; protect or enter on breaks.

3

Bracket / OCO / OTO

Entry with attached take-profit and stop-loss.

4

day · gtc · opg · cls · ioc · fok

How long an order stays working.

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K3

Performance and Risk Metrics

1

Return

Total return · CAGR = (end/start)^(1/years) − 1.

2

Risk

Volatility = σ(daily) × √252 · maximum drawdown.

3

Risk-adjusted

Sharpe · Sortino · Calmar = CAGR / |MaxDD|.

4

Trade quality

Win rate · profit factor · expectancy · turnover · exposure.

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Classification of Trading Algorithms by Purpose

	Purpose	Examples
Execution algorithms	Work a large order at low cost and market impact	VWAP, TWAP, POV, implementation shortfall, iceberg, smart order routing
Market making	Quote bids and offers, earn the spread, manage inventory	Passive market making with resting limit orders
Arbitrage	Capture mispricing between related products or markets	ETF versus its basket, index futures versus stocks
Directional / alpha	Take a view on the direction of price	Trend following, momentum, breakout, mean reversion, news
Statistical arbitrage and pairs	Hedged: long one asset, short a related one; trade the spread	Pairs trading, cointegration, market-neutral baskets
Factor and portfolio	Build and rebalance portfolios from return drivers and risk	Value, momentum, size factors; mean-variance optimisation
Hedging and risk	Offset unwanted exposure and enforce limits	Delta hedging, beta hedging, pairs hedge, kill switch

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SEC Classification of High-Frequency Strategies (2010)

1

Passive market making

Submits non-marketable resting bids and offers that provide liquidity at specified prices

2

Arbitrage

Captures pricing inefficiencies between related products or markets, e.g. an ETF and its basket

3

Structural

Exploits structural vulnerabilities in the market or its participants, e.g. stale quotes via faster data

4

Directional

Takes a short-term position anticipating an intraday price move in a particular direction

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Classification by Holding Period

	Horizon	Typical algorithms
High-frequency	Microseconds to seconds	Market making, ETF and latency arbitrage
Intraday	Minutes to hours	VWAP / TWAP execution, intraday momentum
Swing	Days to weeks	Mean reversion, pairs trading
Position	Weeks to months	Trend following, factor portfolios

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Classification by Model Type

1. **Rule-based**
   Fixed if-then rules on indicators, e.g. a moving-average crossover

2. **Statistical**
   Econometric tests and models: cointegration, regression, t-tests

3. **Machine learning**
   Models learn from labelled data: logistic regression, random forest, XGBoost

4. **Reinforcement learning**
   Learns a policy from rewards, e.g. order-execution research

5. **LLM and agentic**
   Language models read text and agents plan, code and act under limits

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Strategy Families at a Glance

1

Trend following

Rides sustained moves in the direction of the trend; cuts losers quickly - e.g. Buy when the 50-day moving average crosses above the 200-day

2

Mean reversion

Bets that prices stretched far from their average will snap back - e.g. Buy when price closes below the lower Bollinger Band (20-day, 2 std dev)

3

Momentum and breakout

Buys recent winners or prices breaking out of a range, expecting them to keep going - e.g. Buy on a close above the 20-day high (Donchian channel breakout)

4

Statistical / pairs arbitrage

Trades the spread between two related assets when it drifts from its normal level - e.g. Short the rich leg, buy the cheap leg when the spread z-score exceeds 2

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Strategy Families at a Glance

1

Execution algorithms

Split a large order to minimise cost and market impact: VWAP, TWAP, POV, implementation shortfall - e.g. POV: trade 10% of market volume until the order is filled

2

Market making

Continuously quotes buy and sell prices, earning the spread while managing inventory - e.g. Quote bid and ask around mid-price; skew quotes as inventory builds

3

Index rebalancing / event-driven

Trades predictable flows around index changes, earnings and corporate events - e.g. Position ahead of announced index inclusions on the rebalance date

4

Sentiment / NLP

Turns news, filings and social media text into trading signals using NLP or LLMs - e.g. Go long when the average FinBERT headline sentiment is strongly positive

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TOPIC 2 · HOW THE ALGORITHMS WORK

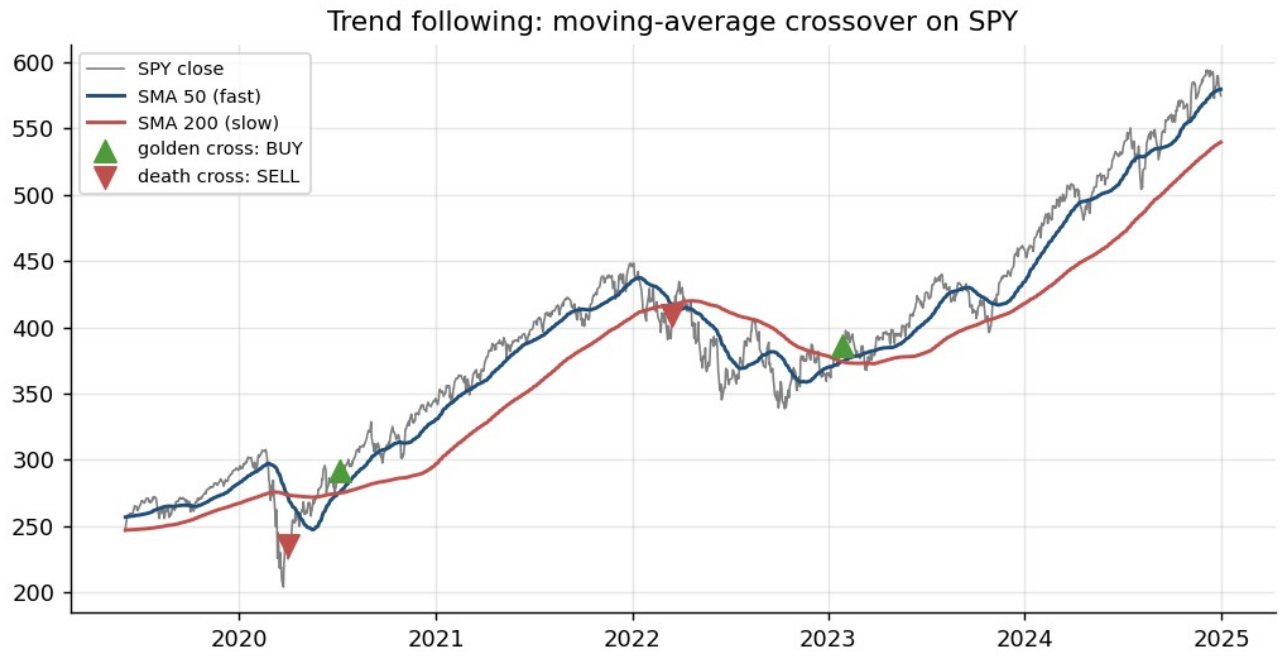
Concepts over code

Every algorithm below is coded for you - by the simulator or an AI assistant. Learn what it does, when it works and what can go wrong; let AI handle the syntax.

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Algorithm: Moving-Average Crossover (Trend Following)



HOW IT WORKS

Two moving averages of the close: a fast one reacts quickly, a slow one shows the trend.

RULE

Fast crosses ABOVE slow (golden cross): buy. Fast crosses BELOW (death cross): sell or go flat.

WORKS BEST WHEN

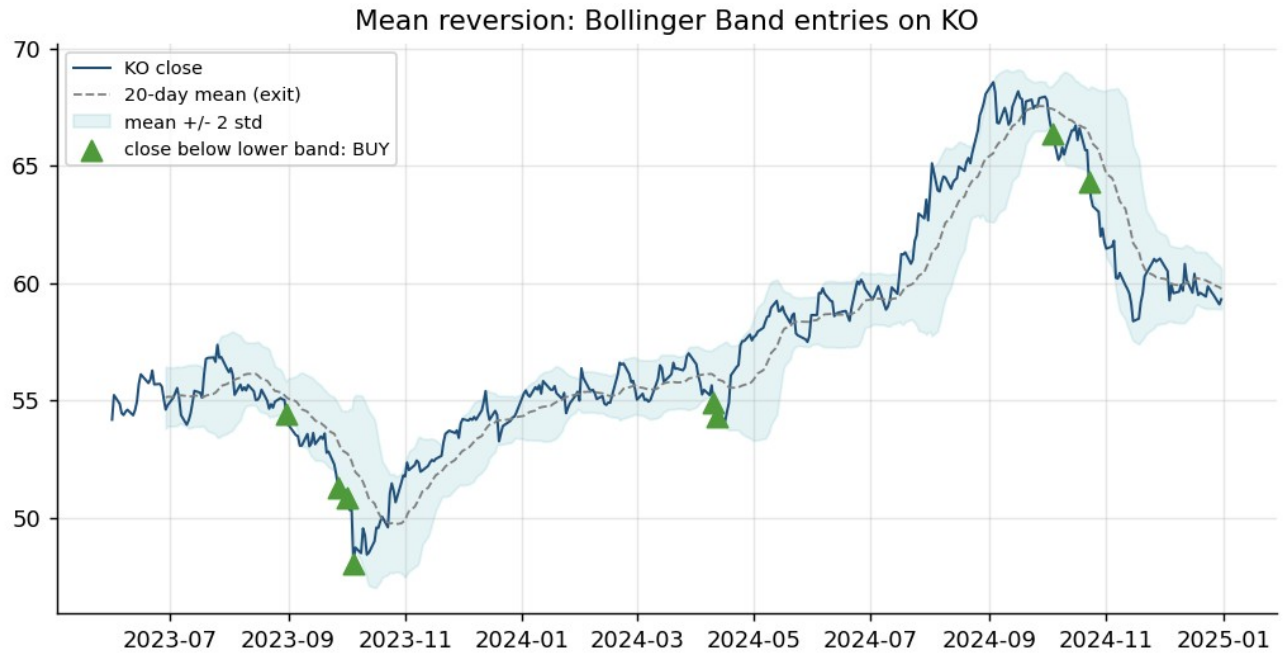
Markets trend for months - it rides the move and cuts losers when the trend turns.

WATCH OUT FOR

Whipsaws in sideways markets; signals lag; trade on the next bar, not the crossing bar.

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Algorithm: Bollinger Band Mean Reversion



HOW IT WORKS

Bands = 20-day mean +/- 2 standard deviations. A close outside the band is 'stretched'.

RULE

Close below the lower band: buy. Exit when the price returns to the middle band (the mean).

WORKS BEST WHEN

Range-bound markets where over-reactions snap back, e.g. a defensive stock such as KO.

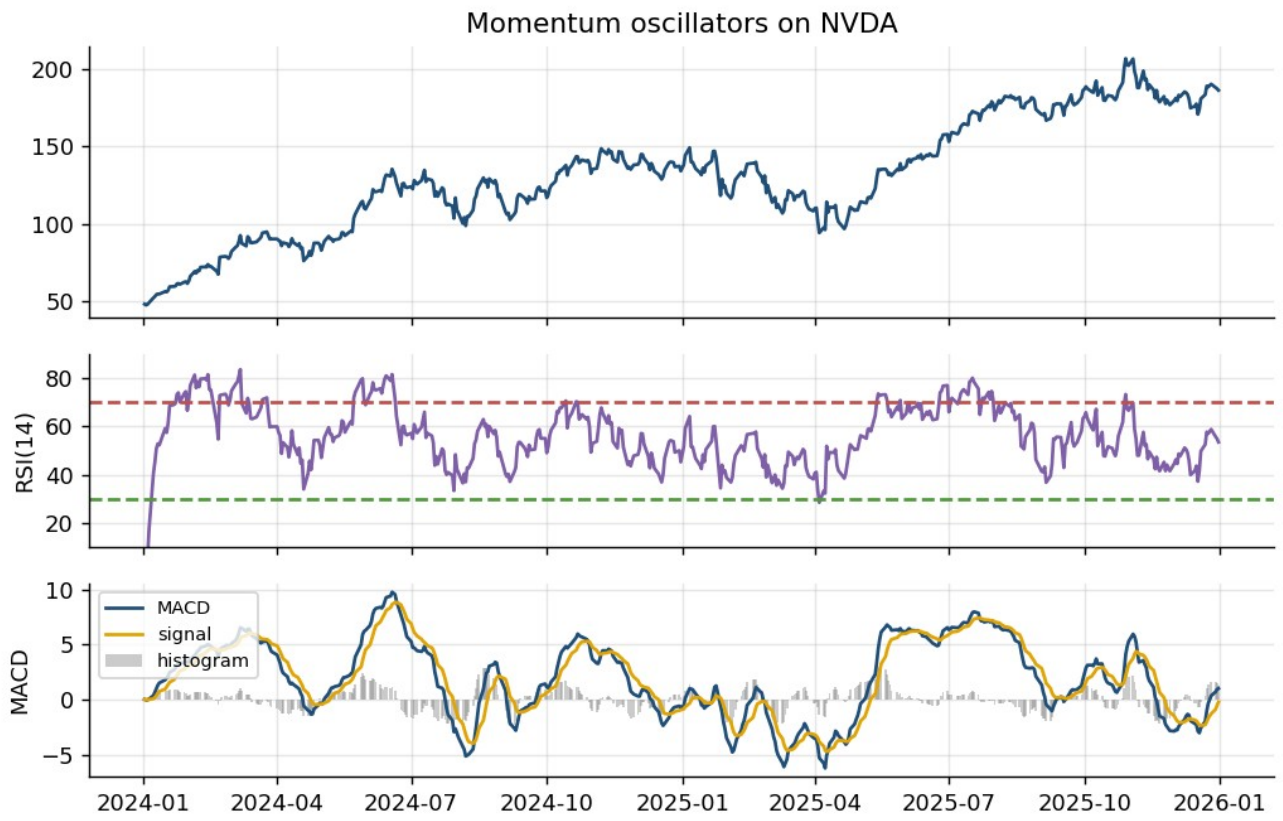
WATCH OUT FOR

Strong trends keep 'stretching' - add a stop-loss or a trend filter; many small wins, rare big losses.

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Algorithm: RSI and MACD Momentum Oscillators



HOW IT WORKS

RSI = 100 - 100/(1+RS), RS = average gain / average loss over 14 bars. MACD = EMA12 - EMA26.

RULE

RSI < 30 oversold / > 70 overbought. MACD crossing above its 9-EMA signal line = momentum buy.

WORKS BEST WHEN

As filters or confirmation for a trend or mean-reversion rule, not as stand-alone predictors.

WATCH OUT FOR

Oscillators can stay overbought in strong trends; RSI and MACD are highly correlated with price.

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Algorithm: Cross-Sectional Momentum



HOW IT WORKS

Rank a universe by past return (e.g. 12 months, skipping the last month) each month.

RULE

Hold the top-ranked assets, avoid or short the bottom-ranked; rebalance monthly.

WORKS BEST WHEN

Information spreads slowly and winners keep winning for several months.

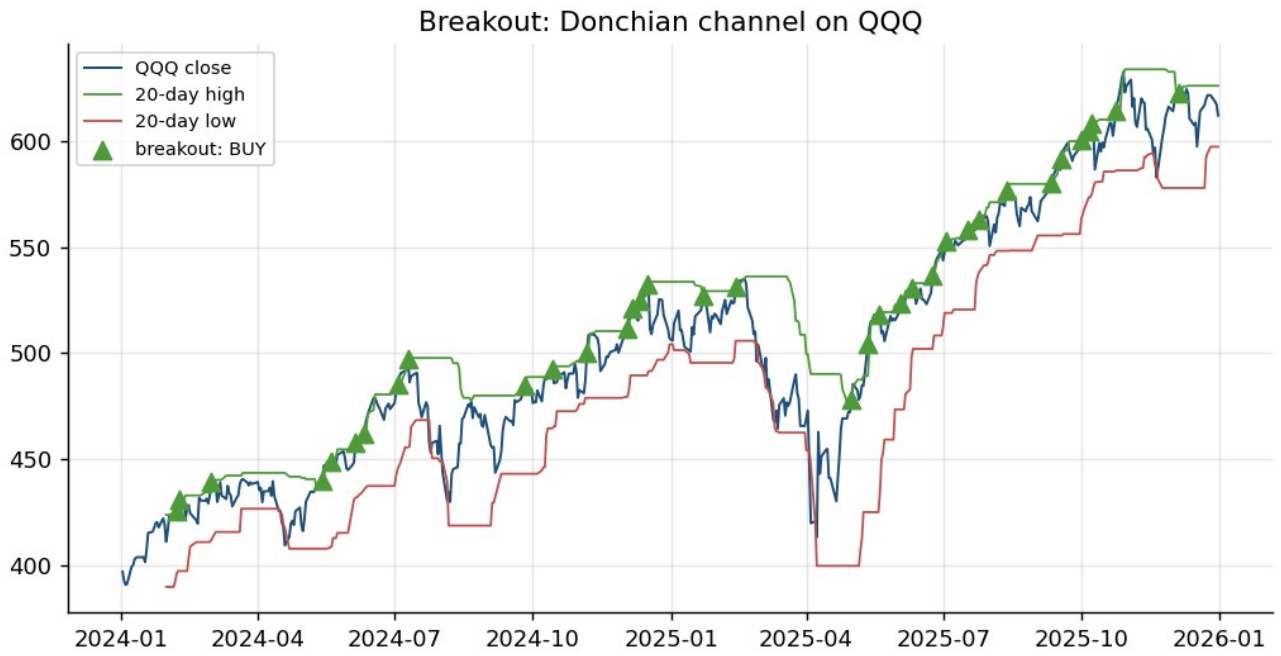
WATCH OUT FOR

Sharp 'momentum crashes' when markets reverse; turnover costs at every rebalance.

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Algorithm: Donchian Channel Breakout



HOW IT WORKS

Track the highest high and lowest low of the last n bars (n = 20 here).

RULE

Close above the 20-day high: buy the breakout. Exit below the 20-day low (or a trailing stop).

WORKS BEST WHEN

New trends start with a break of a range - the rule used by the 1980s 'Turtle Traders'.

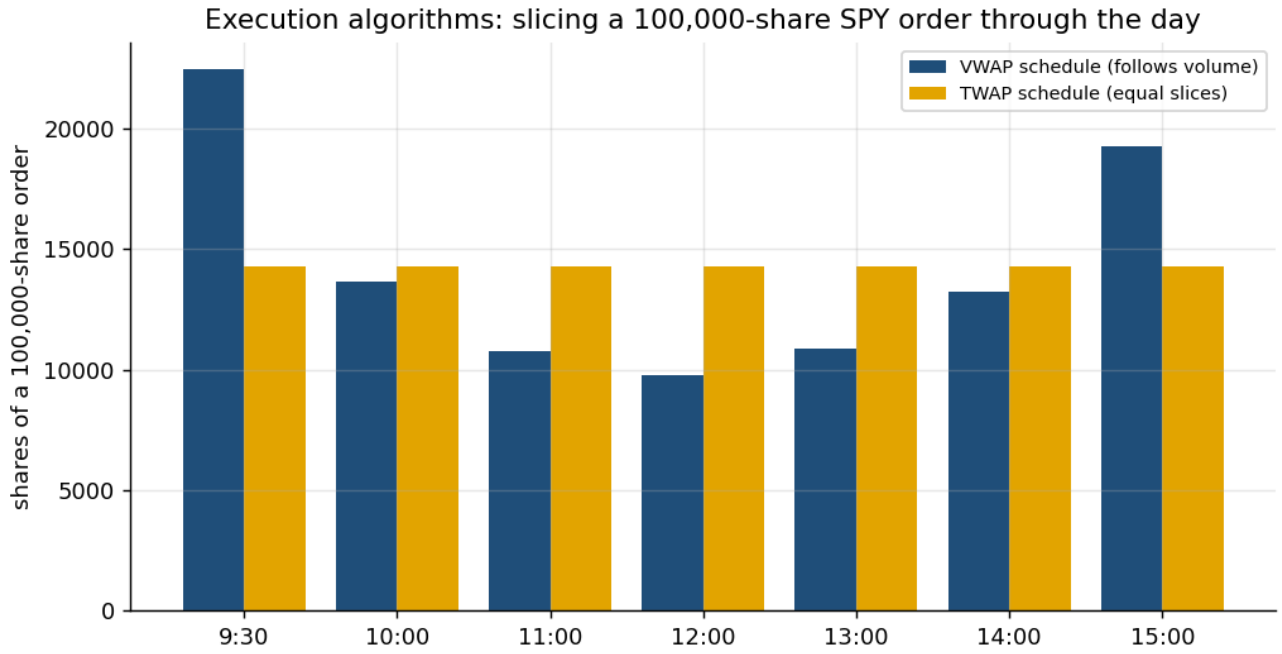
WATCH OUT FOR

False breakouts in choppy markets; the channel must use PREVIOUS bars only (no look-ahead).

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Algorithm: VWAP and TWAP Execution



HOW IT WORKS

Split a large parent order into child orders through the day to reduce market impact.

RULE

TWAP: equal slices per time interval. VWAP: slices proportional to the typical volume curve.

WORKS BEST WHEN

Working client or fund orders too large to trade at once; benchmarked to the day's VWAP.

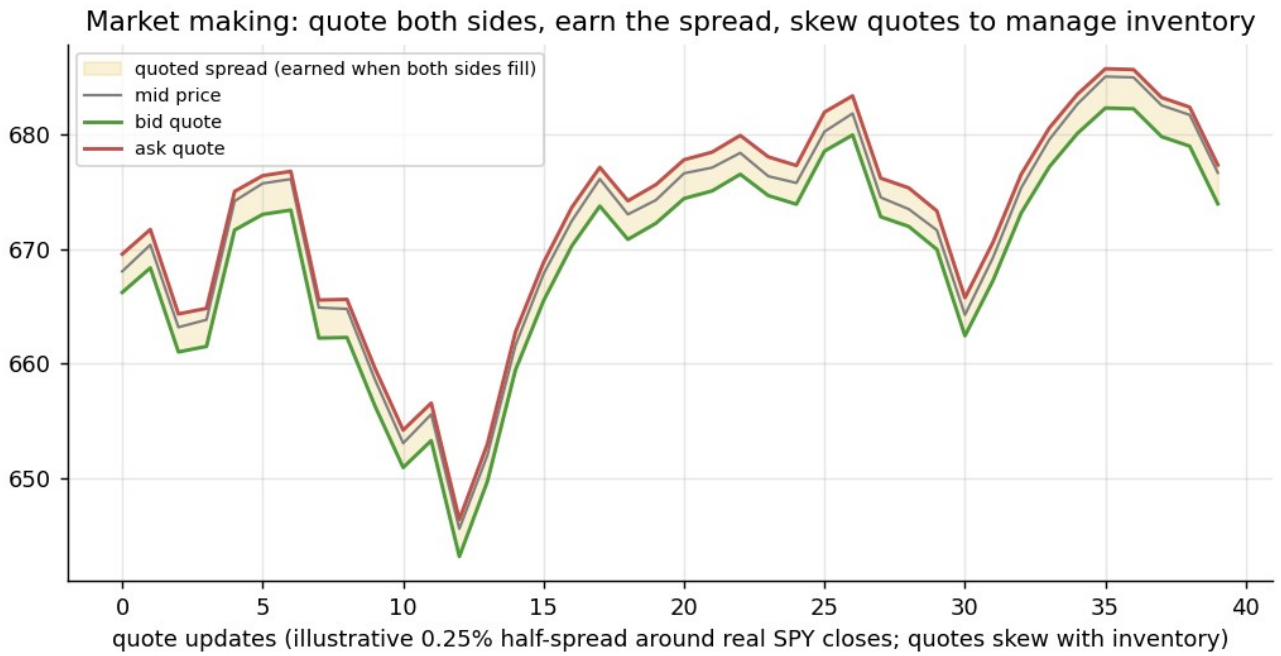
WATCH OUT FOR

VWAP relies on a volume forecast; both are predictable to other traders - add randomisation.

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K3

Algorithm: Market Making



HOW IT WORKS

Quote a bid below and an ask above the mid price at the same time.

RULE

Earn the spread when both sides fill; skew quotes down when long, up when short, to cut inventory.

WORKS BEST WHEN

Liquid markets with steady two-way flow - the business of quant market makers.

WATCH OUT FOR

Adverse selection by better-informed traders; inventory risk in fast moves; needs low latency.

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TradingView versus Alpaca

	TradingView	Alpaca
What it is	Charting and strategy-design platform	API-first US broker for automated trading
Markets	Global charts incl. SGX, US, FX, crypto	US stocks and ETFs, options, crypto
Strategy design	Pine Script v6 indicators and strategies	Write it yourself in Python (alpaca-py)
Backtesting	Strategy Tester (simplified costs and fills)	None built in - you build a cost-aware backtest
Paper trading	Built-in Paper Trading account from the chart	Paper account mirroring the live API
Automation	Alerts; webhooks on paid plans	Trading API: orders, positions, clock; 24/5 trading
Data	Charts and Data Window	Market Data API: bars, quotes, trades, news (IEX free)
Machine learning	Not supported	Any Python library (scikit-learn, XGBoost)
Best used for	Designing, visualising and first-testing a rule	Executing and forward-testing it automatically

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Paper Trading on TradingView, Step by Step

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Supercharts > Trade > Paper Trading > Connect
2. Set starting capital and commission
3. Order ticket: side, quantity, Market / Limit / Stop
4. Add stop loss and take profit; place order
5. Track Positions, Orders, Account History
6. Close the position; reset when done

Paper fills use TradingView's prices but cannot show your own market impact - compare them with the backtest's assumed costs.

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Paper Trading on Alpaca, Step by Step - Manual and Automated

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Dashboard: Paper account > search SPY > Buy
2. Watch Orders: new > accepted > filled
3. Outside US hours: extended-hours LIMIT orders
4. python -m pipeline execute (order manager)
5. python -m pipeline run --loop --live-paper
6. Check dashboard; kill switch to stop

From Singapore the US regular session is 21:30-04:00 (22:30-05:00 in northern winter); class-time orders use Alpaca's 24/5 extended-hours limit orders.

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The Order Lifecycle

A program must read the status — never assume a fill.

1. new
2. accepted
3. partially filled
4. filled
5. cancelled / expired / rejected

Bracket orders protect a position even if your program stops running.

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From Idea to Tested Rule on TradingView

1. Chart the instrument and timeframe
2. Add indicators (Pine)
3. Write the strategy() script
4. Set commission and slippage
5. Read the Strategy Tester
6. Compare with buy-and-hold

The Strategy Tester is a first filter; the full test with costs runs in Python.

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A3 K3 K7

Indicator Stage: Pine Script v6 Indicators and their Python Twins

LAB 05

Plot SMA/EMA, Bollinger Bands, RSI and MACD and export a Pine Script v6 strategy for TradingView. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Overlay trend indicators
2. Simulator: Oscillators
3. Simulator: Export Pine Script v6
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

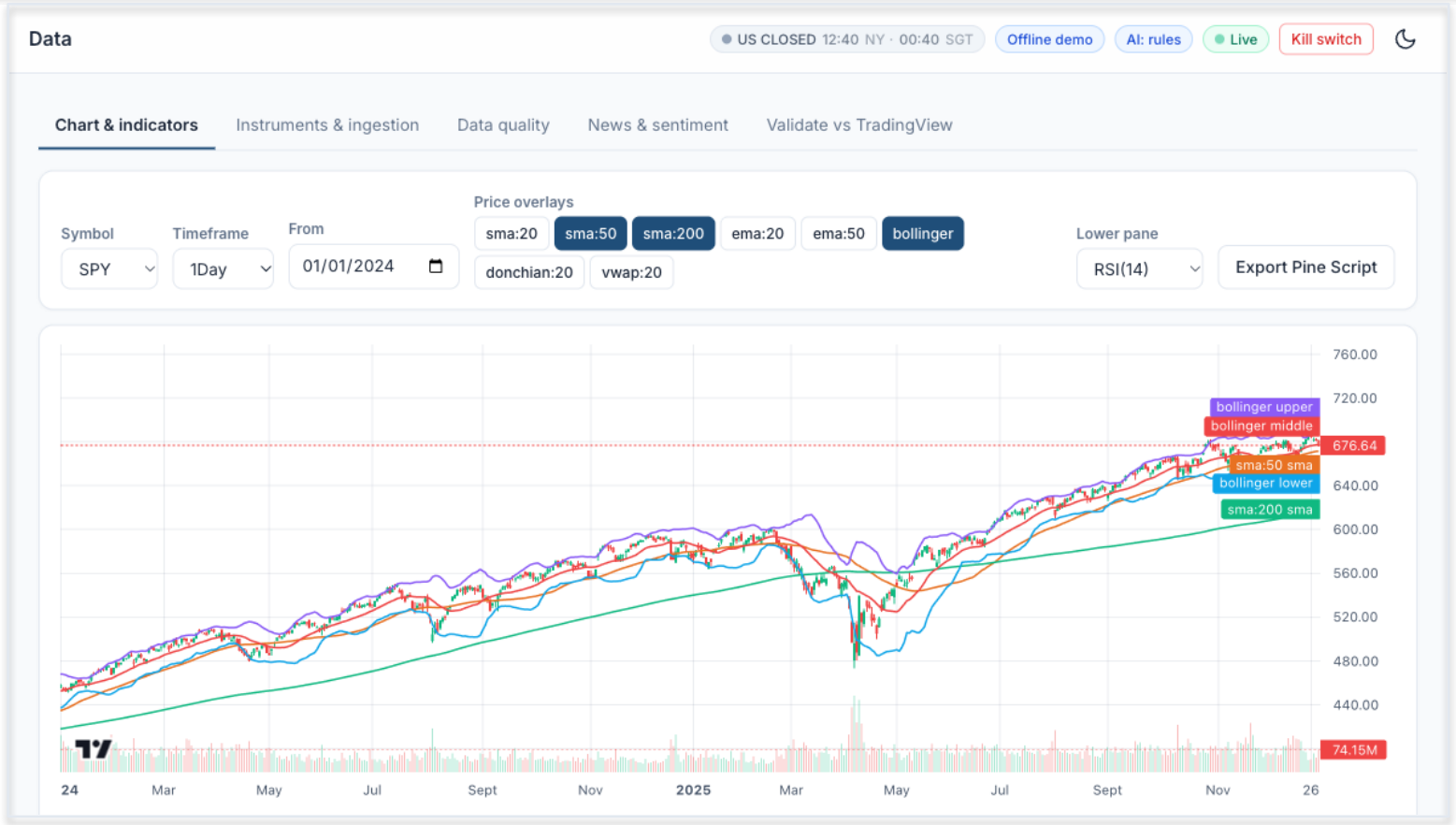
Pine Script v6 indicators on TradingView and their Python twins, cross-checked value by value.

Duration
25 min

© Tertiary Infotech Academy Pte Ltd

LAB 05 · SIMULATOR

Lab 05 — In the Trading Simulator



Course Labs > Lab 05 at http://localhost:8000 - Data > Chart SPY with SMA 50/200 and Bollinger. Offline demo data; connect Alpaca paper keys for live paper trading.

© Tertiary Infotech Academy Pte Ltd

TOPIC 2 · HANDS-ON LAB

A2 A3 K3 K7

Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

LAB 06

Run the same MA-crossover in TradingView's Strategy Tester and in the app, reconcile the results, and try TradingView Paper Trading. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Backtest the SMA 50/200 crossover in the app
2. Simulator: Run it in TradingView
3. Simulator: Reconcile and paper trade on TradingView
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

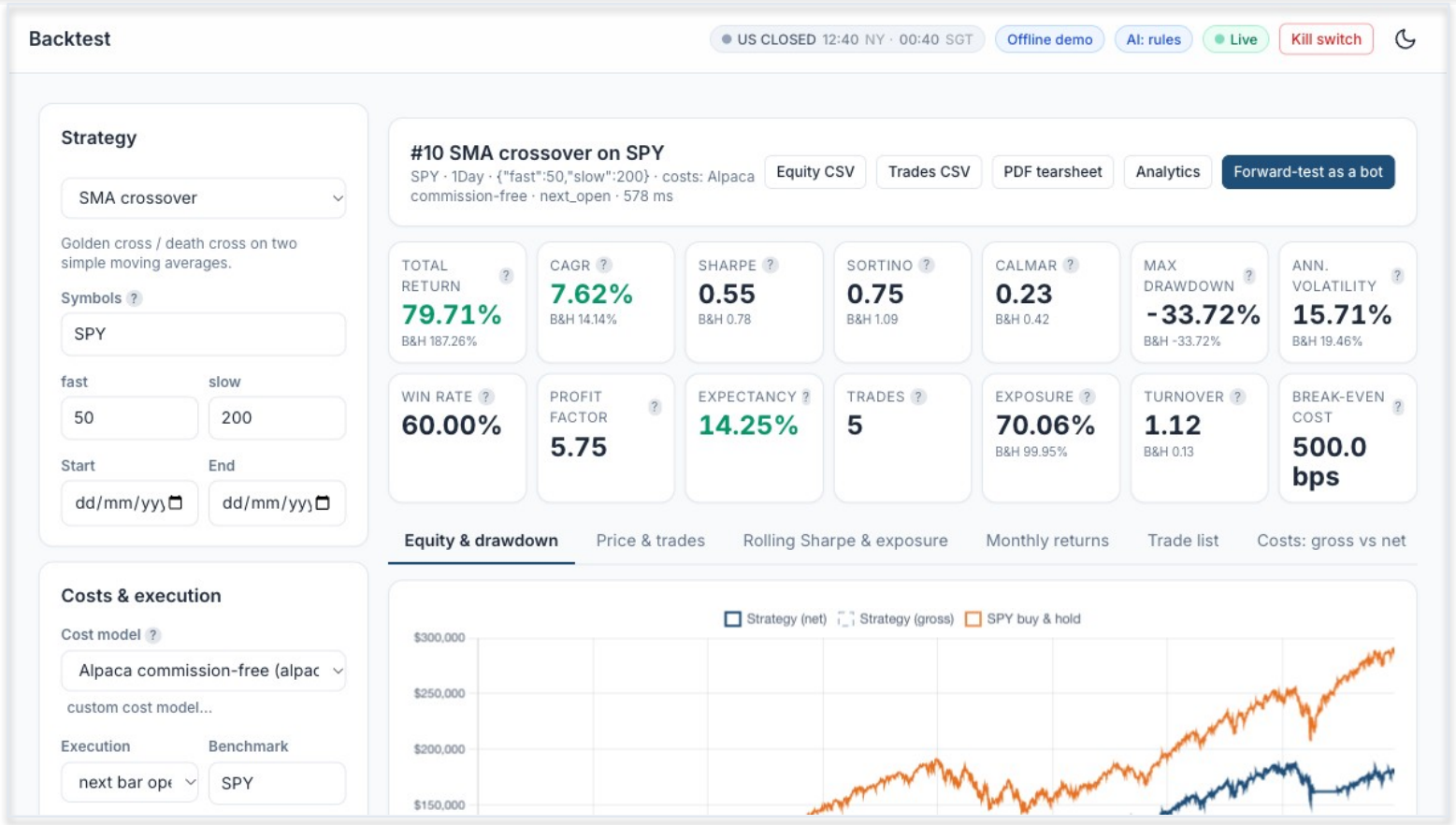
A Pine MA-crossover strategy with costs, reconciled with the pipeline and paper-traded from the chart.

Duration
30 min

© Tertiary Infotech Academy Pte Ltd

LAB 06 · SIMULATOR

Lab 06 — In the Trading Simulator



Course Labs > Lab 06 at http://localhost:8000 - Backtest SPY SMA 50/200, Alpaca costs. Offline demo data; connect Alpaca paper keys for live paper trading.

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Tea Break

15 minutes · see you at 15:40

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A2 K3 K7

Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals

LAB 07

Pick and customise one strategy from each family and compare them after costs. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Trend: EMA 20/50 on SPY
2. Simulator: Mean reversion: Bollinger on KO
3. Simulator: Momentum and breakout
4. Simulator: Read each strategy's explanation
5. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

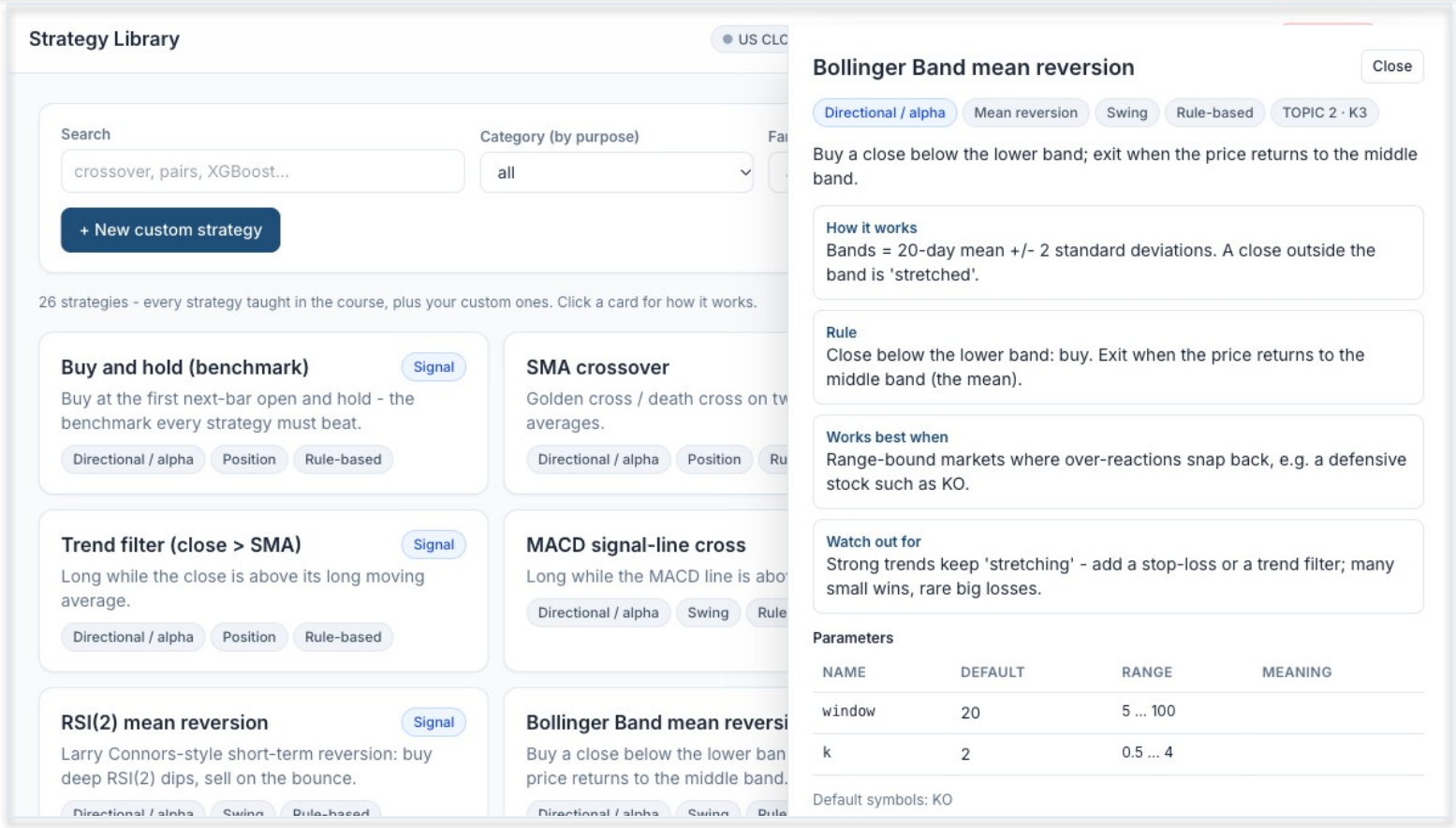
A signal library (trend, mean reversion, momentum, breakout) with next-bar alignment and look-ahead tests.

Duration
25 min

© Tertiary Infotech Academy Pte Ltd

LAB 07 · SIMULATOR

Lab 07 — In the Trading Simulator



Course Labs > Lab 07 at http://localhost:8000 - Backtest SPY EMA 20/50, Alpaca costs. Offline demo data; connect Alpaca paper keys for live paper trading.

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Execution Stage: an Order Manager that Trades the Paper Account in Any Session

LAB 08

Place market, limit, stop and bracket paper orders, including an extended-hours limit order, and follow each order's lifecycle. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Market order in the regular session
2. Simulator: Extended-hours limit order (Singapore class hours)
3. Simulator: Bracket order
4. Simulator: Queue while closed and cancel
5. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

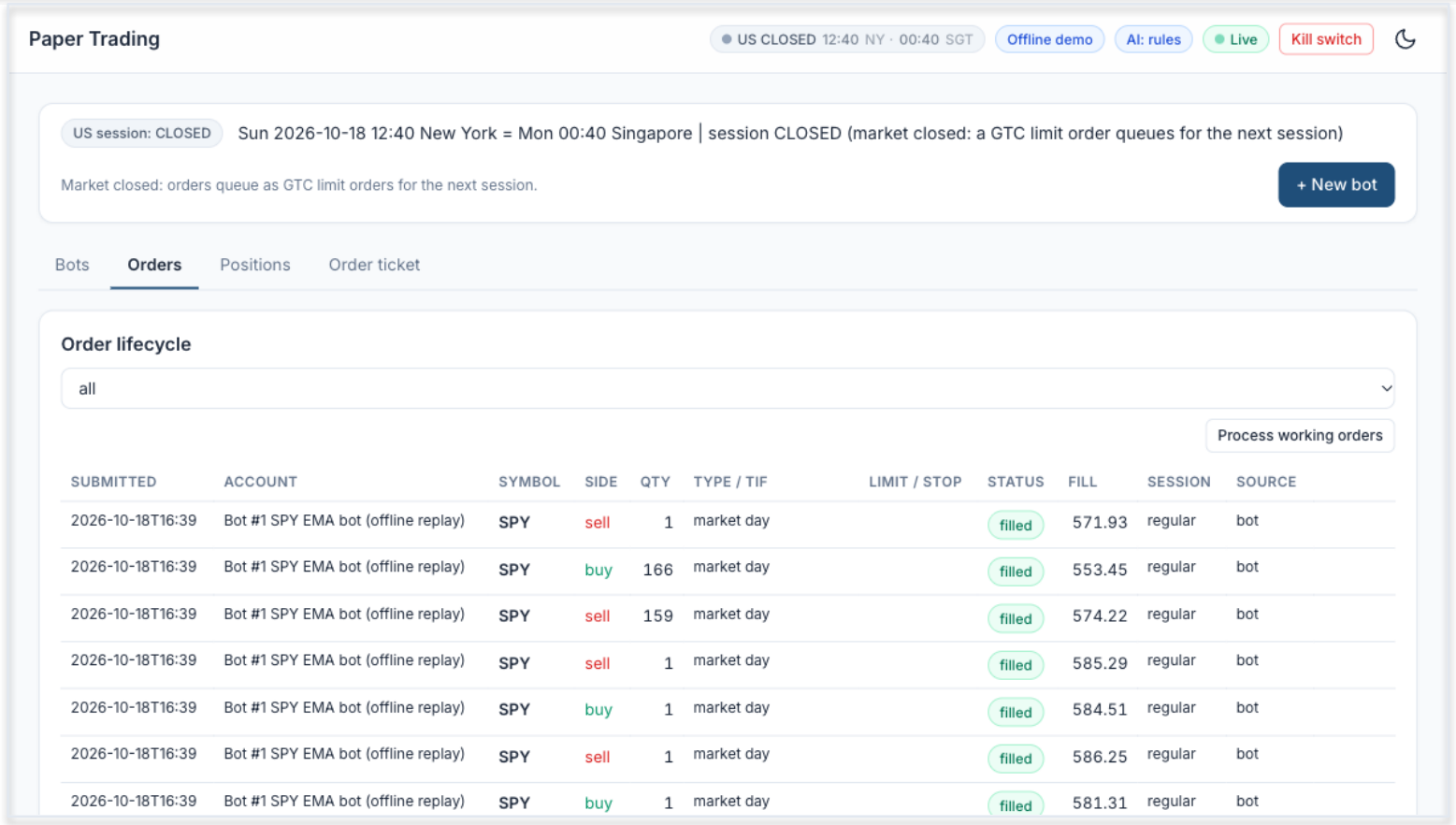
An order manager that turns target positions into paper orders in regular or extended hours.

Duration
30 min

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LAB 08 · SIMULATOR

Lab 08 — In the Trading Simulator



Course Labs > Lab 08 at http://localhost:8000 - Paper Trading > Order ticket. Offline demo data; connect Alpaca paper keys for live paper trading.

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A3 K3

Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

LAB 09

Read the full metrics table and compare a strategy with buy-and-hold. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Open the analytics
2. Simulator: Compare with buy-and-hold
3. Simulator: Glossary
4. Code: rebuild the stage with your AI assistant

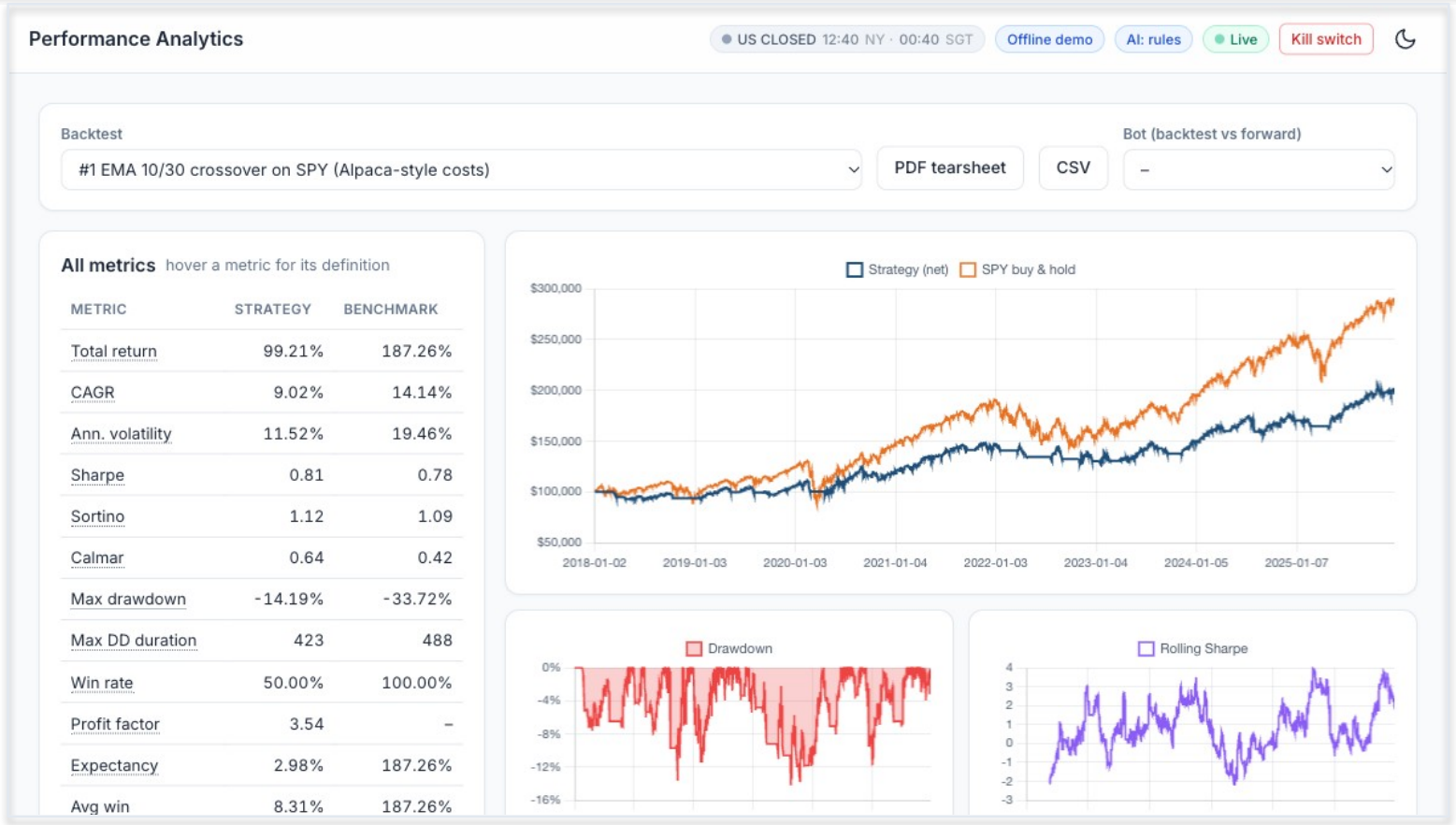
YOU'LL PRODUCE

An evaluation stage producing the metrics and benchmark table for every strategy.

Duration
25 min

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Lab 09 — In the Trading Simulator



Course Labs > Lab 09 at http://localhost:8000 - Analytics. Offline demo data; connect Alpaca paper keys for live paper trading.

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A2 A3 K3 K5 K7

Topic 2 Recap

1. **A2**
   Identify appropriate statistical algorithms and data models to test hypotheses and/or theories

2. **A3**
   Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements

3. **K3**
   Range and application of various statistical methods and algorithms

4. **K5**
   Usage of analytics platforms and tools

5. **K7**
   Coding languages for programming of algorithms and signals

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End of Day 1

See you at 09:00 tomorrow

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TOPIC 3

03

ML Models and A/B Testing

LEARNING OUTCOME

Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.

Abilities A4, A5 · Knowledge K2, K4, K6 · Day 2 · Labs 10–14 · Assessed by PP Task 3 and WA Q2, Q4, Q6

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K2

Types of Statistical Analyses

1

Descriptive

What happened — mean return, volatility, skew, correlation.

2

Inferential

Is it real? — t-test, bootstrap, confidence intervals.

3

Predictive

What next? — regression, classification.

4

Prescriptive

What to do? — optimisation, position sizing.

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K2

Families of Learning Algorithms

1

Supervised

Features → known target: up/down, next return.

2

Unsupervised

Structure without a target: regimes, PCA factors.

3

Reinforcement

Learn a policy from rewards — execution research.

4

Search & simulation

Grid search, Monte Carlo, bootstrap, walk-forward.

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Supervised Models for Trading Signals

1

Linear / logistic regression

Interpretable baselines; P(up) from a weighted sum.

2

Decision tree & random forest

Threshold splits; averaged trees reduce variance.

3

XGBoost

Sequential boosted trees — strongest on tabular features.

4

Neural networks

MLP, LSTM, Transformer — data-hungry, easy to overfit.

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Unsupervised Models for Regimes

1

K-means

Minimise within-cluster variance; choose k by elbow or silhouette.

2

Hierarchical clustering

Dendrogram of similar assets for diversification.

3

PCA

First component ≈ the market; yield curve = level, slope, curvature.

4

Fit on the past only

Assign new days to regimes as they arrive.

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Statistical Modelling Techniques

1

Returns, not prices

Closer to stationary; avoids spurious regression.

2

Time-ordered validation

TimeSeriesSplit and walk-forward — never shuffle.

3

Leak-free features

Lags and rolling windows known at decision time.

4

Regularise

L1/L2, depth limits, early stopping, fewer features.

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A/B Testing Two Strategy Variants

1

State H0 and H1 first

H0: equal mean returns; H1: B is higher. α = 5%.

2

Welch's t-test

Compare mean daily returns with unequal variances.

3

Bootstrap the Sharpe difference

Resample returns; 95% CI excluding 0?

4

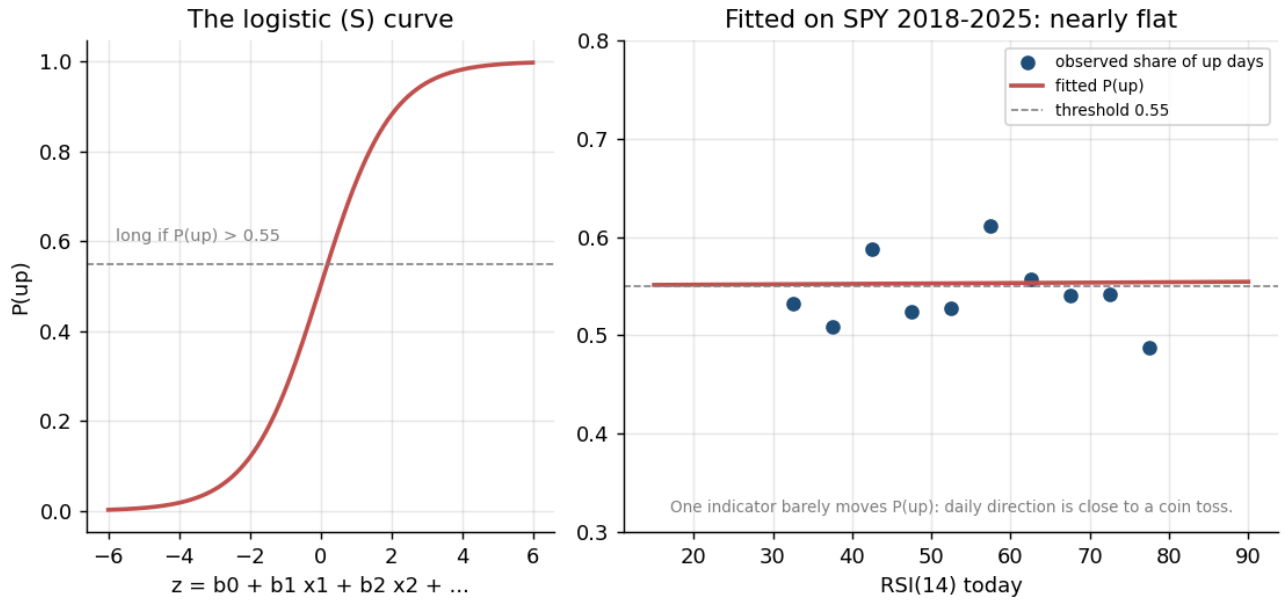
Mind multiple testing

100 variants → ~5 false 'winners' by chance.

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K4

Algorithm: Logistic Regression for Direction



HOW IT WORKS

Fits P(up tomorrow) = 1 / (1 + e^-(b0 + b1 x1 + ...)) from features known today.

RULE

Go long when P(up) exceeds a threshold such as 0.55; otherwise stay flat.

WORKS BEST WHEN

A transparent baseline: each coefficient shows how a feature moves the probability.

WATCH OUT FOR

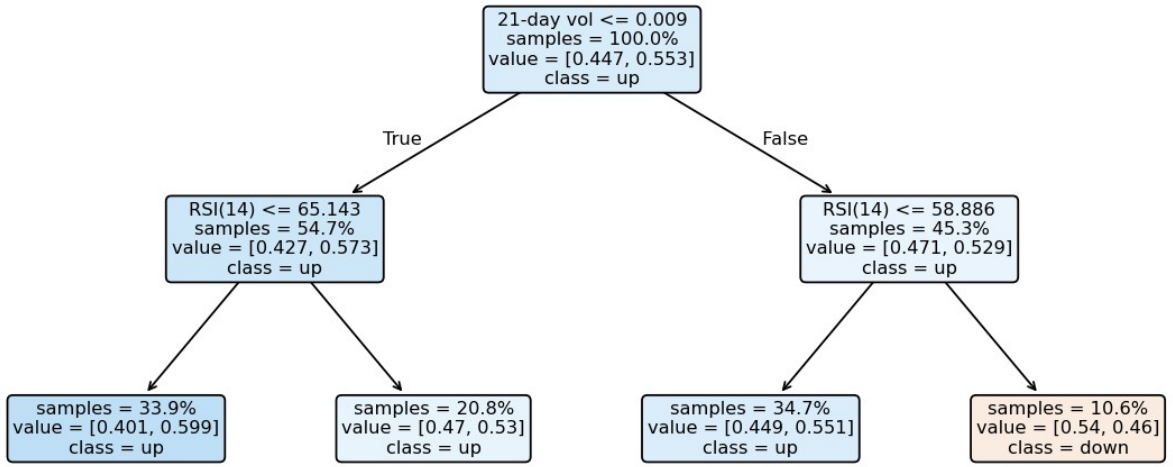
Only linear effects unless you engineer features; daily direction is close to a coin toss.

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K4

Algorithm: Decision Tree and Random Forest

Decision tree (depth 2) on SPY: split on feature thresholds; a random forest averages many such trees



HOW IT WORKS

A tree splits the data on feature thresholds (e.g. RSI < 30) into leaves with an up/down share.

RULE

A random forest averages hundreds of trees, each grown on a random sample of rows and features.

WORKS BEST WHEN

Non-linear, interacting signals - e.g. oversold AND low volatility.

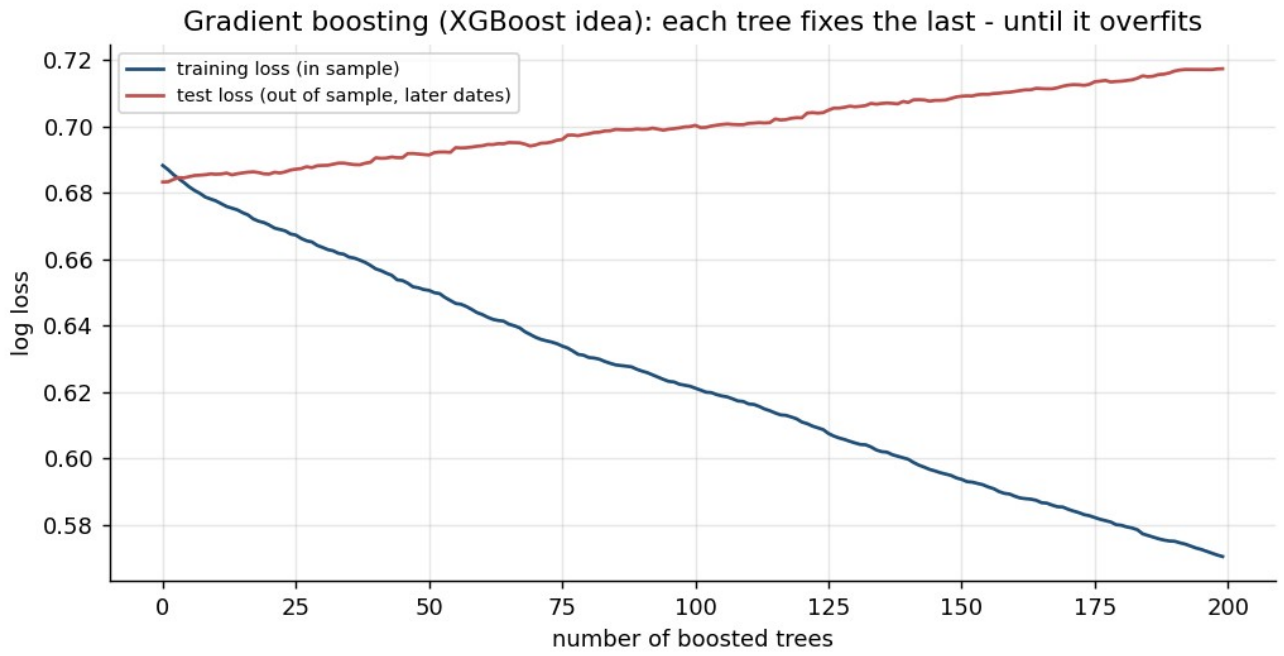
WATCH OUT FOR

Deep trees memorise noise; limit depth and leaf size and validate walk-forward.

© Tertiary Infotech Academy Pte Ltd

K4

Algorithm: Gradient Boosting (XGBoost)



HOW IT WORKS

Builds trees one after another; each new tree corrects the errors of the ensemble so far.

RULE

Prediction = sum of many small trees x learning rate; stop early when test loss stops falling.

WORKS BEST WHEN

Tabular features such as returns, indicators and volatility - usually the strongest model.

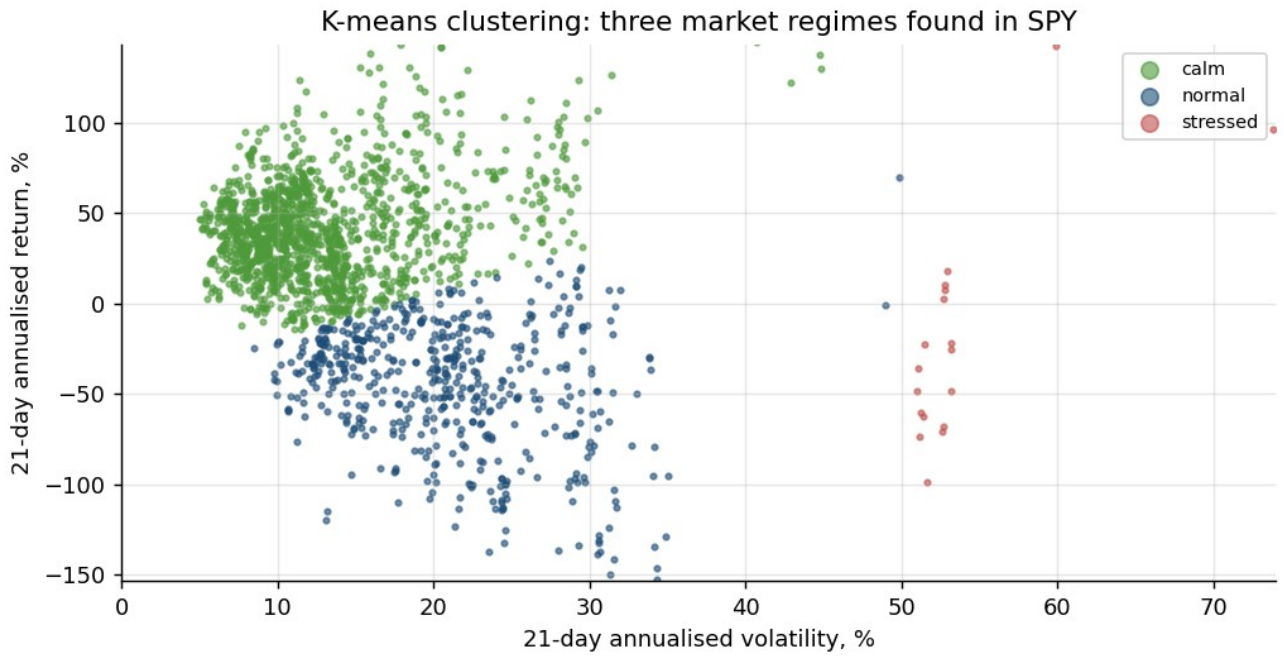
WATCH OUT FOR

Training loss keeps falling while test loss rises: classic overfitting (see the chart).

© Tertiary Infotech Academy Pte Ltd

K4

Algorithm: K-Means Clustering for Market Regimes



HOW IT WORKS

Groups days into k clusters by minimising the distance of each day to its cluster centre.

RULE

Features: rolling return and volatility (standardised). Name clusters by their averages.

WORKS BEST WHEN

Switching or sizing strategies by regime: trend in calm markets, cut risk when stressed.

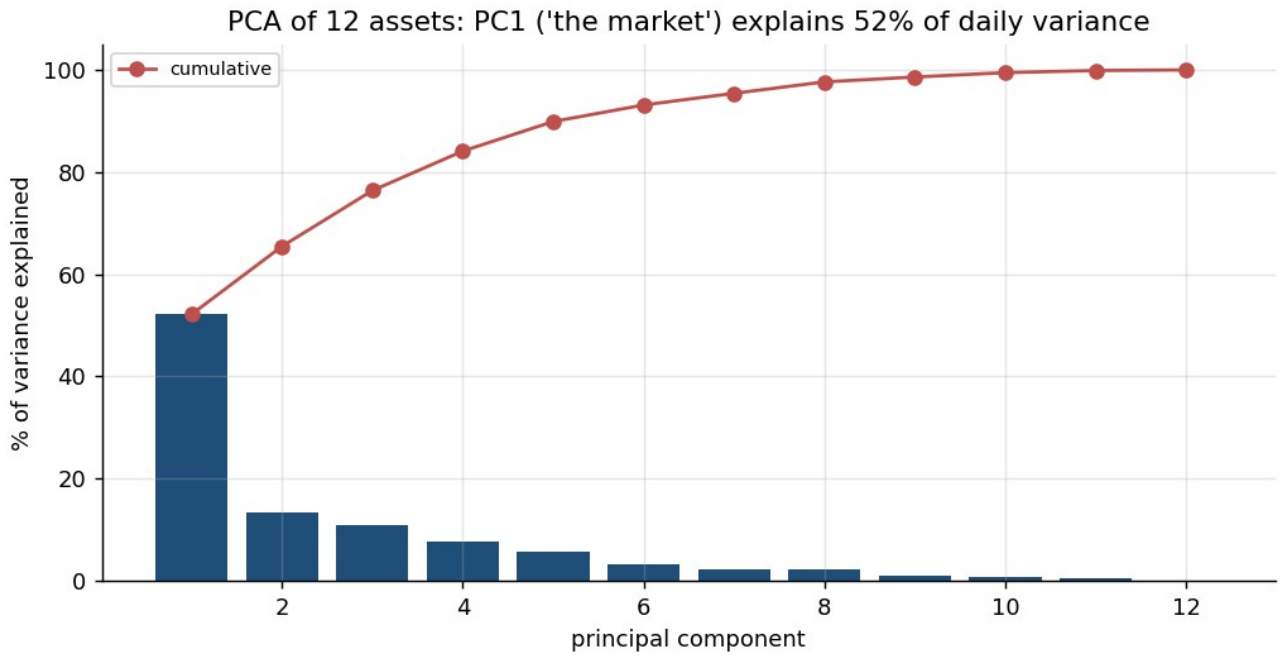
WATCH OUT FOR

Fit on past data only; choose k with the elbow or silhouette method; regimes shift.

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K4

Algorithm: Principal Component Analysis (PCA)



HOW IT WORKS

Finds uncorrelated combinations of assets that explain the most variance of returns.

RULE

PC1 of a stock universe is 'the market'; later PCs are sector or style factors.

WORKS BEST WHEN

Reducing many correlated features; building factor-neutral and pairs portfolios.

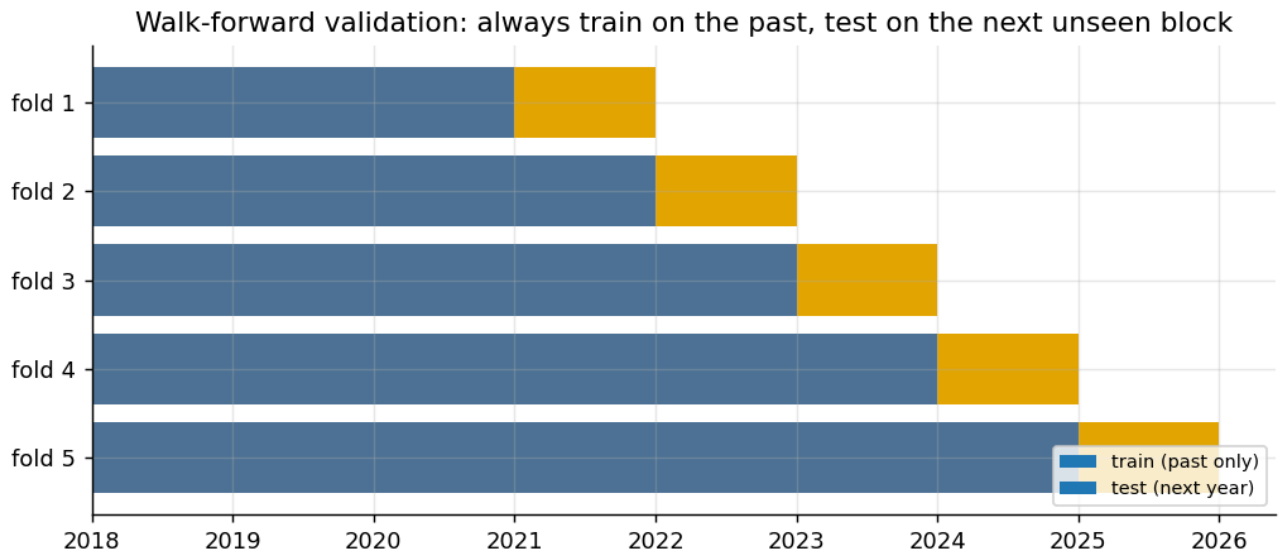
WATCH OUT FOR

Standardise first; components change over time; PCs are statistical, not economic, factors.

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K6

Method: Walk-Forward Validation



HOW IT WORKS

Train on a past window, test on the next unseen block, then roll forward and repeat.

RULE

Never shuffle time series; each test block is genuinely out of sample.

WORKS BEST WHEN

Estimating how a model or parameter set would really have performed live.

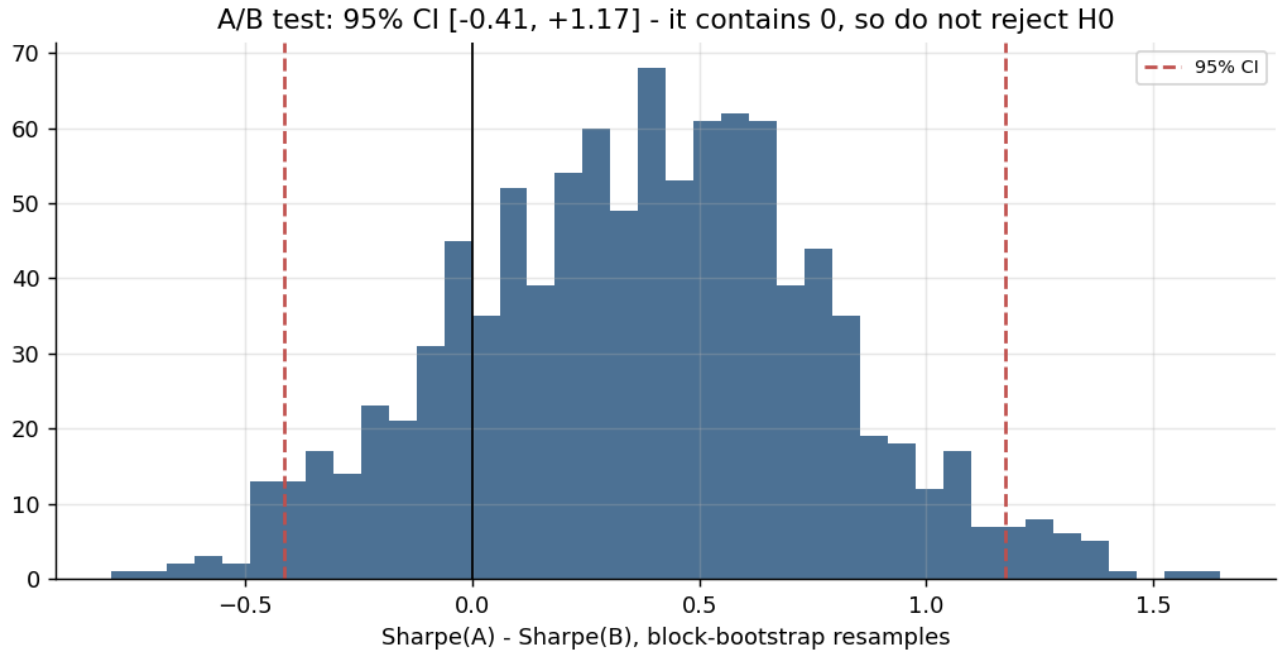
WATCH OUT FOR

Overlapping targets leak information - leave a gap between train and test.

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A5

Method: A/B Testing Two Strategy Variants



HOW IT WORKS

Compare variant A and B on the same days: H0 'no difference' vs H1 'B is better'.

RULE

Welch / paired t-test on daily returns; bootstrap a 95% CI of the Sharpe difference.

WORKS BEST WHEN

Deciding which parameter set or filter to promote to paper trading.

WATCH OUT FOR

If the CI contains 0, do not reject H0; testing 100 variants gives ~5 false winners.

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From Probability to Position

1. Build leak-free features
2. Train on past window
3. Predict P(up) for tomorrow
4. Long if P(up) > 0.55, else flat
5. Backtest net of costs

Judge the model by the net-of-cost strategy, not by accuracy alone.

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Pairs Trading: a Hedging Algorithm

1

Why it is a hedge

Long the cheap stock, short the rich one in hedge-ratio proportion: market moves largely cancel

2

Choose the pair

Economically linked stocks (e.g. KO and PEP) that are cointegrated, not merely correlated

3

Hedge ratio and spread

Regress log price A on log price B: slope = hedge ratio, residual = the spread (Engle-Granger)

4

Test the spread

ADF test on the residual: a stationary spread tends to revert to its mean

5

Trade the z-score

Enter when $|z| > 2$, exit as z returns towards 0; trade both legs on the next bar

6

Risks

Cointegration breaks, hedge ratio drifts, double costs, short-borrow, multiple testing

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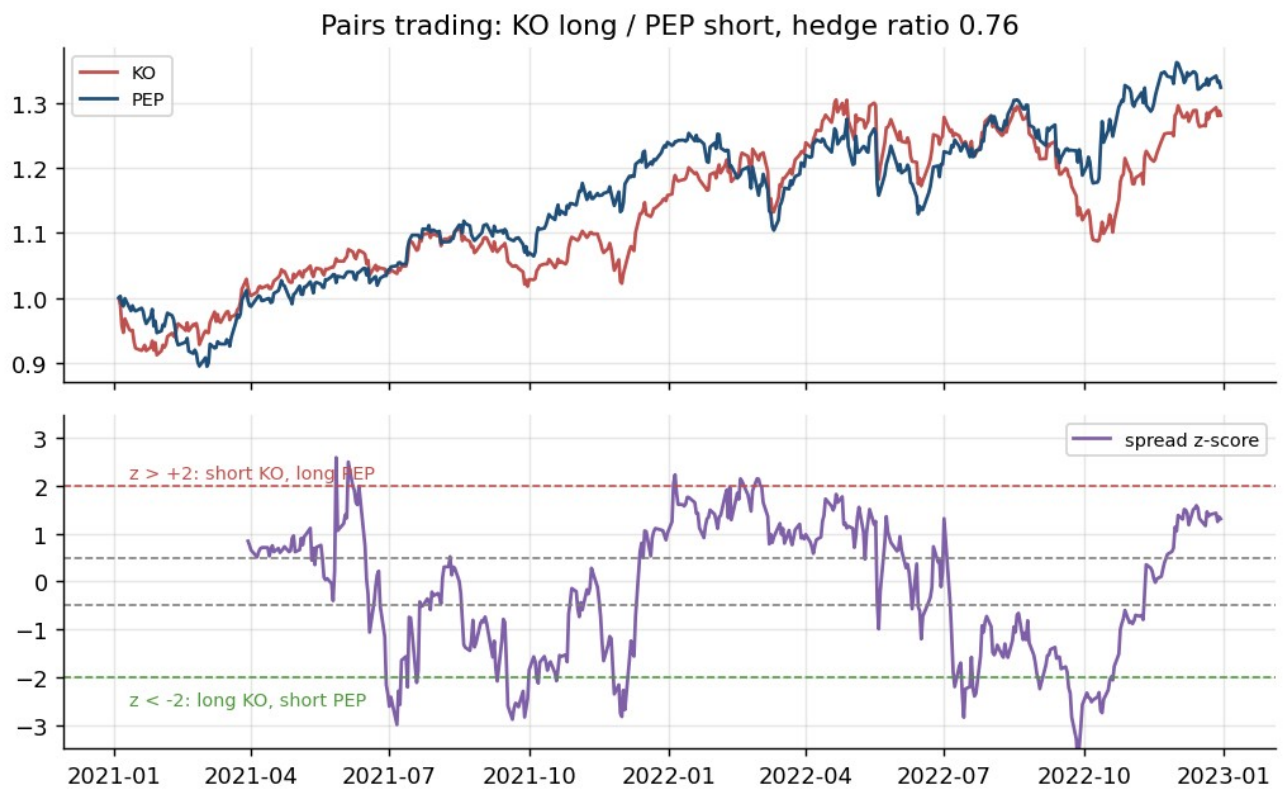
The Pairs-Trading Algorithm Step by Step

1. Select an economically linked pair
2. Estimate the hedge ratio (OLS)
3. Test cointegration (ADF on residual)
4. Compute the spread z-score
5. Enter / exit both legs
6. Monitor and re-test

Gatev, Goetzmann and Rouwenhorst (2006): a simple pairs rule on US stocks earned average annualised excess returns of up to 11% (1962-2002); later studies find smaller and less consistent profits.

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Algorithm: Pairs Trading (Hedged Statistical Arbitrage)



HOW IT WORKS

Long one stock, short a cointegrated partner in hedge-ratio proportion: market moves cancel.

RULE

Spread z > +2: short A / long B. z < -2: long A / short B. Exit as z returns to 0.

WORKS BEST WHEN

Economically linked stocks (KO/PEP) whose spread has a stable mean.

WATCH OUT FOR

Cointegration can break; costs on both legs; short-borrow; screening many pairs.

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A2 A4 K2

Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

LAB 10

Grid-search SMA crossover parameters in-sample and measure how the ranking decays out of sample. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Run the grid
2. Simulator: Read the decay
3. Simulator: Walk-forward
4. Code: rebuild the stage with your AI assistant

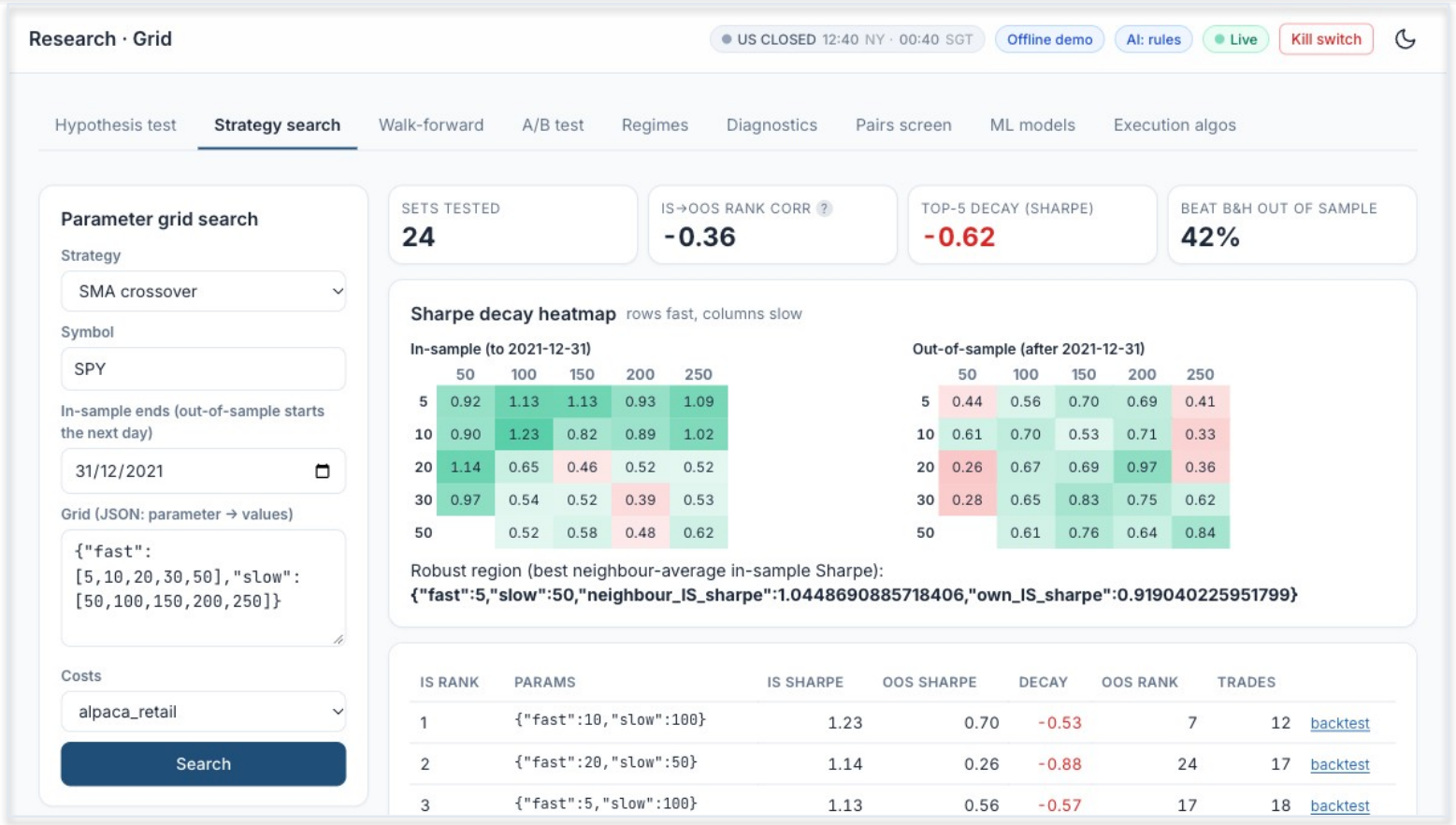
YOU'LL PRODUCE

A strategy-search stage: grid search, out-of-sample ranking and the decay report.

Duration
20 min

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Lab 10 — In the Trading Simulator



Course Labs > Lab 10 at http://localhost:8000 - Research > Grid SPY SMA cross. Offline demo data; connect Alpaca paper keys for live paper trading.

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A4 K4 K6

ML-Signal Stage: a Walk-Forward Classifier from Probability to Position

LAB 11

Train logistic regression, random forest and XGBoost walk-forward and trade P(up) > threshold. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Train the classifiers
2. Simulator: Backtest an ML strategy
3. Simulator: Check for leakage
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

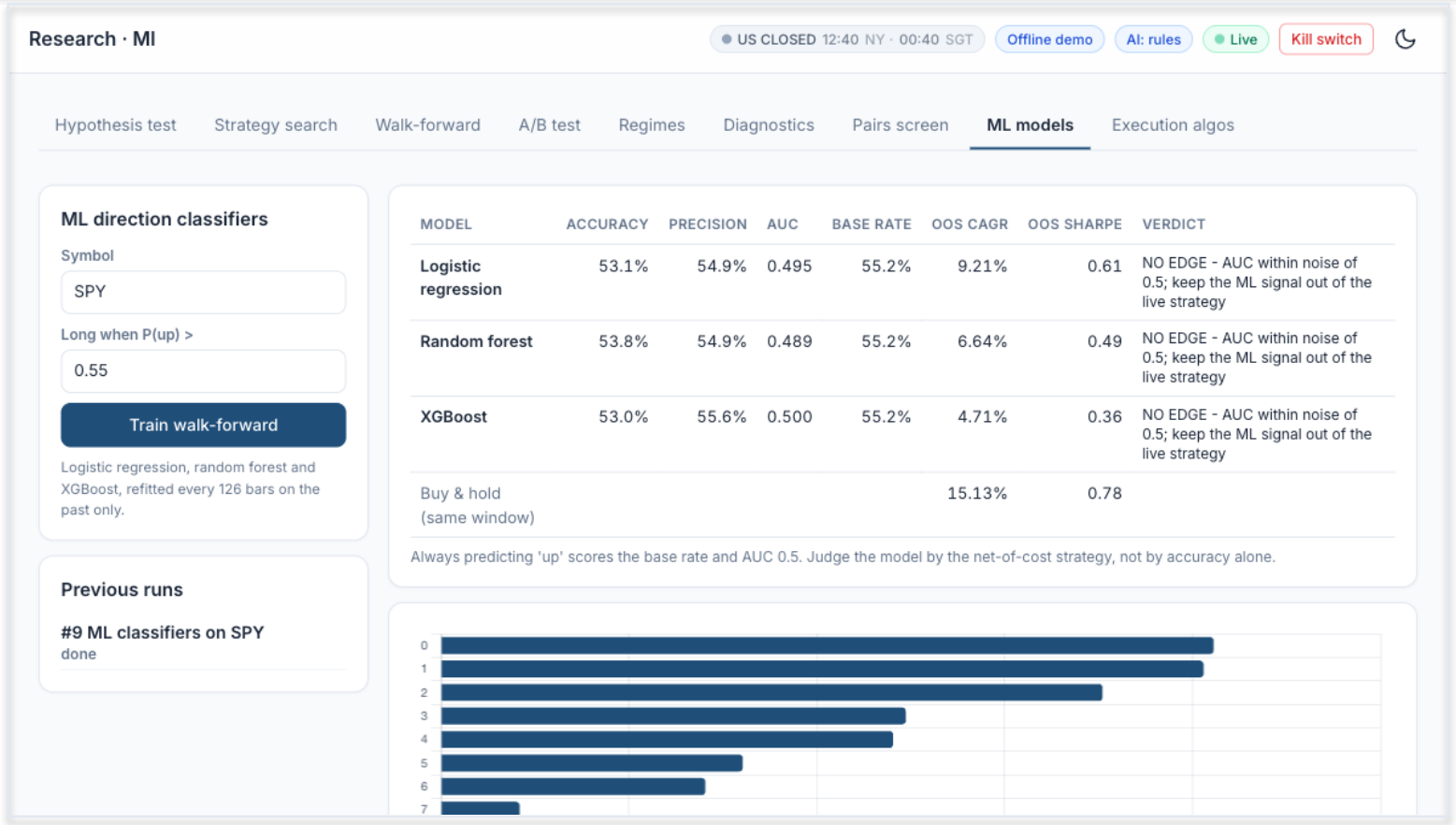
A walk-forward ML signal stage from predicted probability to position, with a leakage check.

Duration
20 min

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LAB 11 · SIMULATOR

Lab 11 — In the Trading Simulator



Course Labs > Lab 11 at http://localhost:8000 - Research > ML SPY. Offline demo data; connect Alpaca paper keys for live paper trading.

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A4 K4 K6

Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

LAB 12

Find market regimes with walk-forward K-means and gate a signal in the high-volatility regime. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Analyse regimes
2. Simulator: Backtest the regime-filtered strategy
3. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

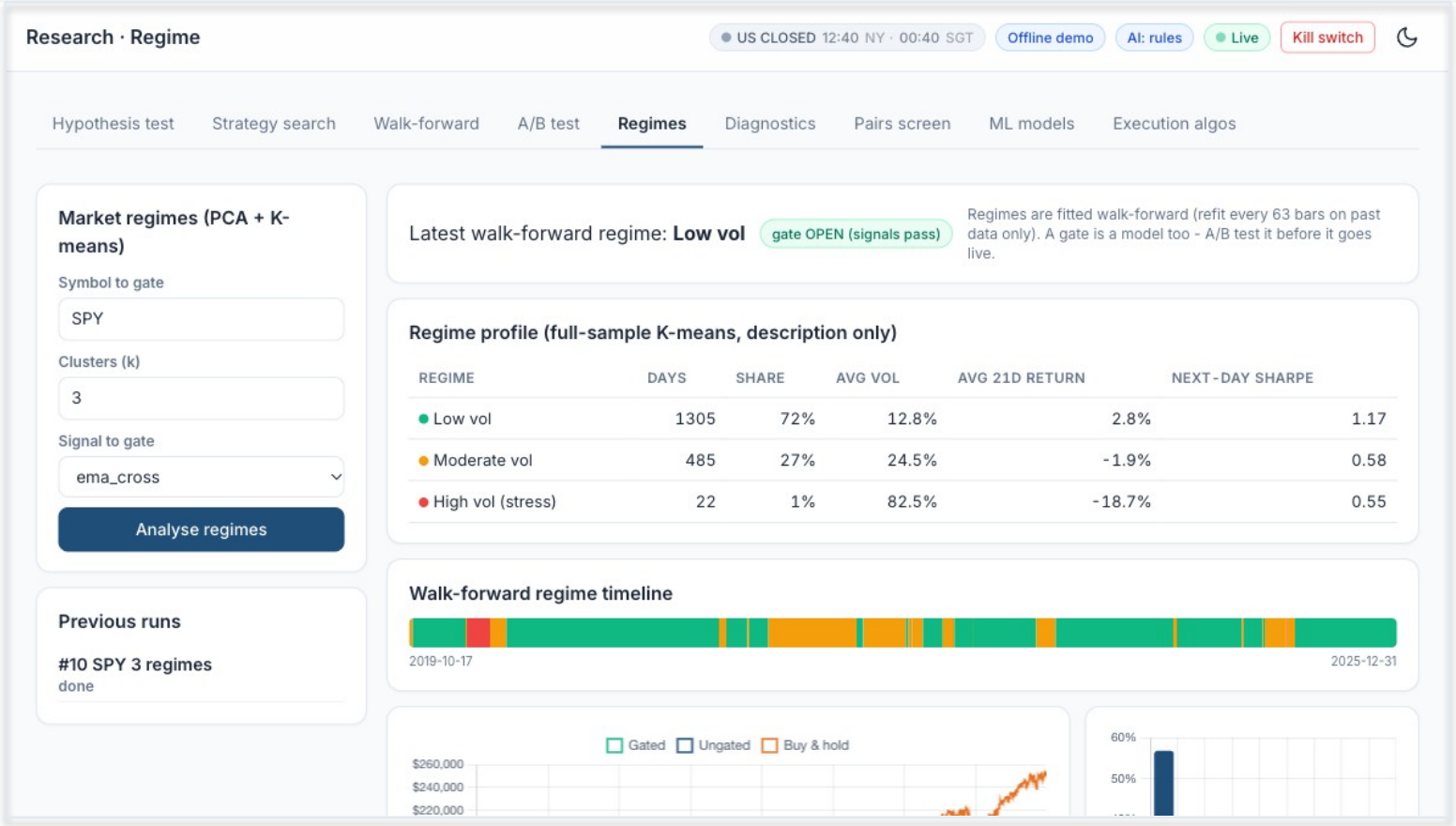
A PCA / K-means regime gate applied to the signals, with before-and-after performance.

Duration
15 min

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LAB 12 · SIMULATOR

Lab 12 — In the Trading Simulator



Course Labs > Lab 12 at http://localhost:8000 - Research > Regimes SPY. Offline demo data; connect Alpaca paper keys for live paper trading.

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A5 K2

A/B-Test Stage: a Statistical Decision before Capital Moves

LAB 13

Decide statistically between two variants before moving capital. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Run the A/B test
2. Simulator: Your own variants
3. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

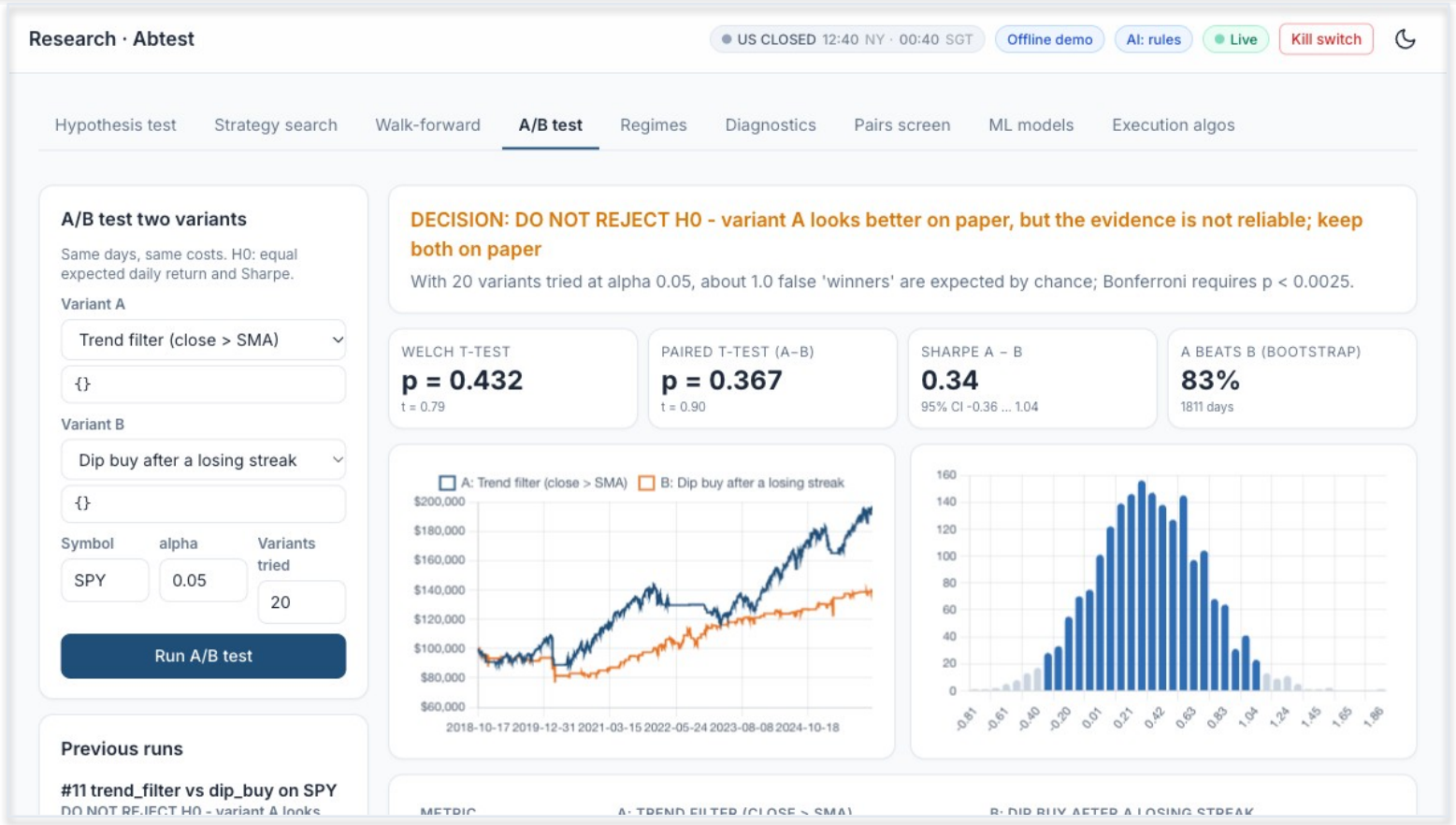
An A/B-test stage that issues a statistical decision before capital moves.

Duration
15 min

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LAB 13 · SIMULATOR

Lab 13 — In the Trading Simulator



Course Labs > Lab 13 at http://localhost:8000 - Research > A/B trend_filter vs dip_buy. Offline demo data; connect Alpaca paper keys for live paper trading.

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A4 K6

Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

LAB 14

Screen pairs for cointegration and backtest a market-neutral KO/PEP spread. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Screen the pairs
2. Simulator: Backtest KO/PEP
3. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

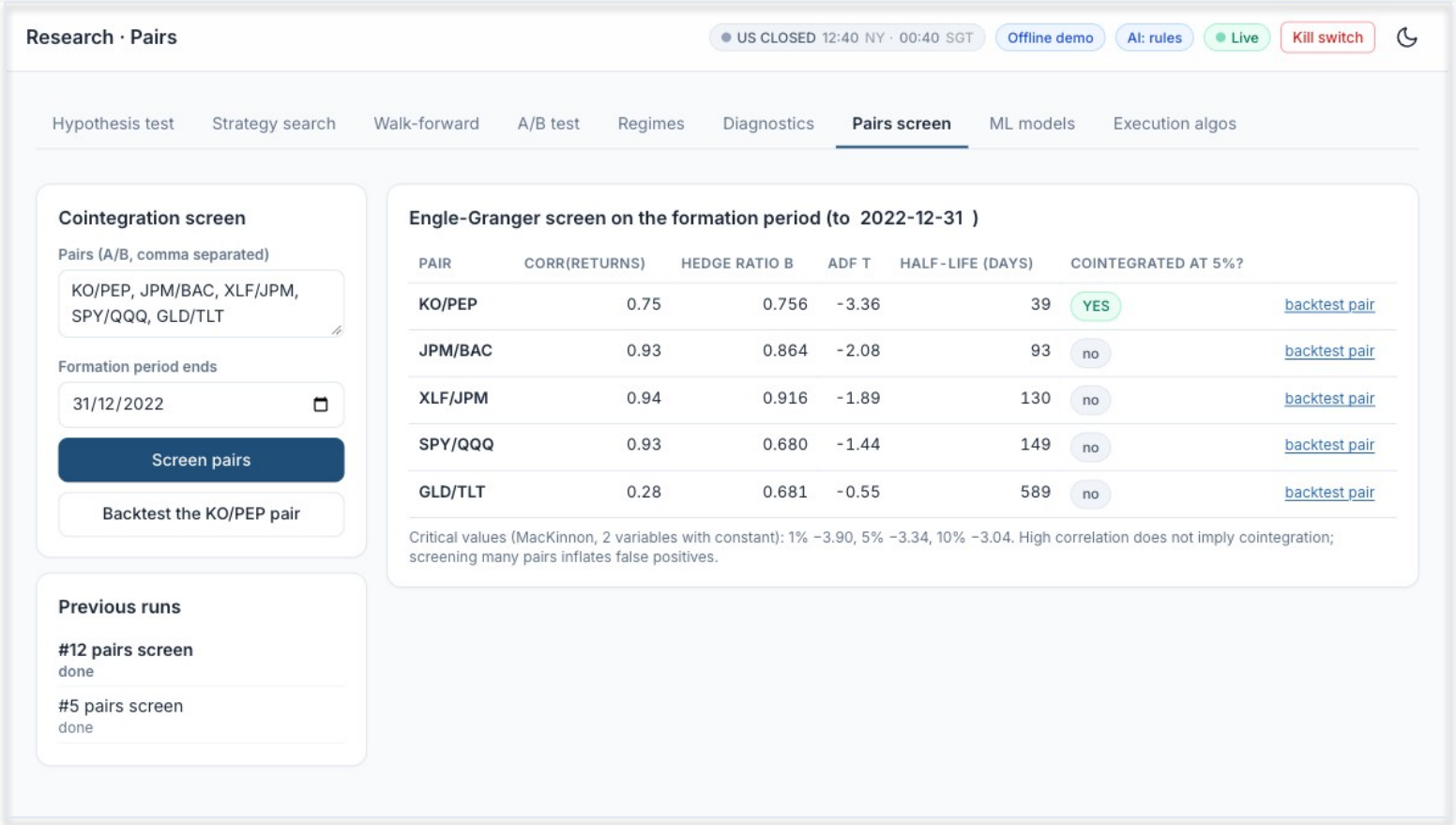
A hedged pairs sleeve: cointegration screen, hedge ratio and a market-neutral spread strategy.

Duration
20 min

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LAB 14 · SIMULATOR

Lab 14 — In the Trading Simulator



Course Labs > Lab 14 at http://localhost:8000 - Research > Pairs screen. Offline demo data; connect Alpaca paper keys for live paper trading.

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A4 A5 K2 K4 K6

Topic 3 Recap

1. **A4**
   Conduct statistical modelling of data to derive patterns and solutions

2. **A5**
   Perform A/B testing based on different hypotheses

3. **K2**
   Types of statistical analyses, data models, algorithms and advanced computational methods

4. **K4**
   Range and application of various types of data models

5. **K6**
   Statistical modelling techniques

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TOPIC 4

04

End-to-End Trading Pipeline

LEARNING OUTCOME

Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.

Abilities A7, A8, A9, A10 · Knowledge K7, K8 · Day 2 · Labs 15–20 · Assessed by PP Task 4 and WA Q7, Q8

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Pre-Trade Risk Controls

1

Position and order limits

Maximum shares, maximum order value per symbol.

2

Maximum daily loss

Stop trading for the day when breached.

3

Market-hours and duplicate checks

Alpaca clock; never resend an open order.

4

Kill switch

One file or flag halts all trading immediately.

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Trading Costs You Must Model

1. **Commission**
   Zero on Alpaca US equities — not on most brokers.

2. **Spread and slippage**
   Half-spread + slippage on every fill, in bps.

3. **Regulatory fees on sells**
   SEC Section 31 fee and FINRA TAF.

4. **Borrow and impact**
   Short-borrow costs; your own market impact.

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Potential Reasons for Unintended Outcomes

1

Overfitting and data snooping

Great in-sample, collapses out of sample.

2

Look-ahead and survivorship bias

Same-bar fills; only today's survivors.

3

Cost and execution drag

Live returns trail the backtest by turnover × cost.

4

Regime change, data and code errors

Drift, bad splits, time zones, duplicate orders.

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What Goes on a Tearsheet

1

Equity versus benchmark

Growth of USD 100,000, strategy and buy-and-hold.

2

Drawdown and rolling Sharpe

Depth, length and stability of the edge.

3

Monthly returns heatmap

When did it make — and lose — money?

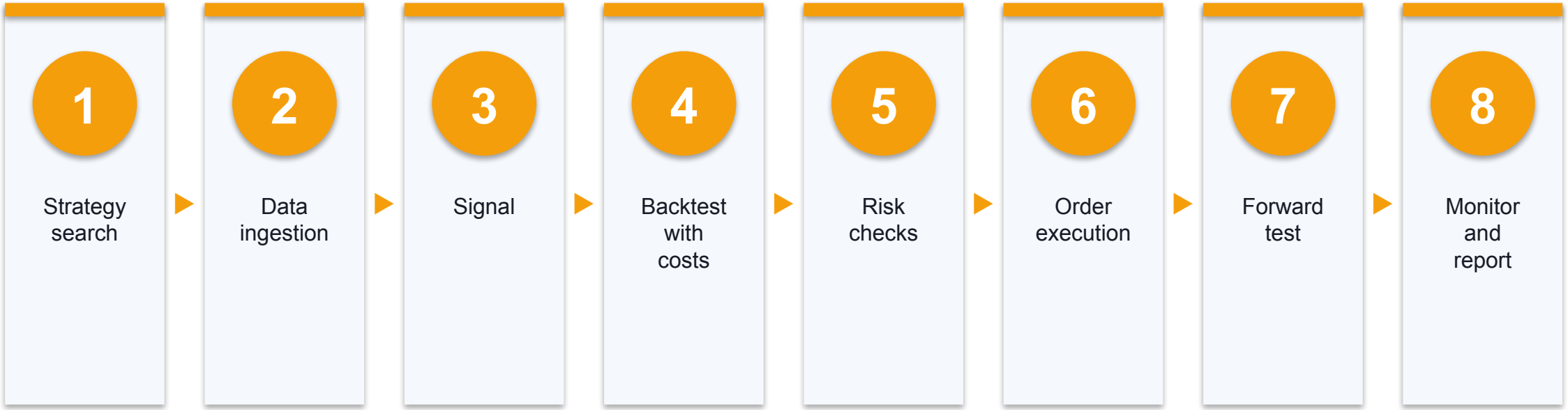
4

Metrics and cost table

Gross versus net; backtest versus forward.

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The End-to-End Algorithmic Trading Pipeline

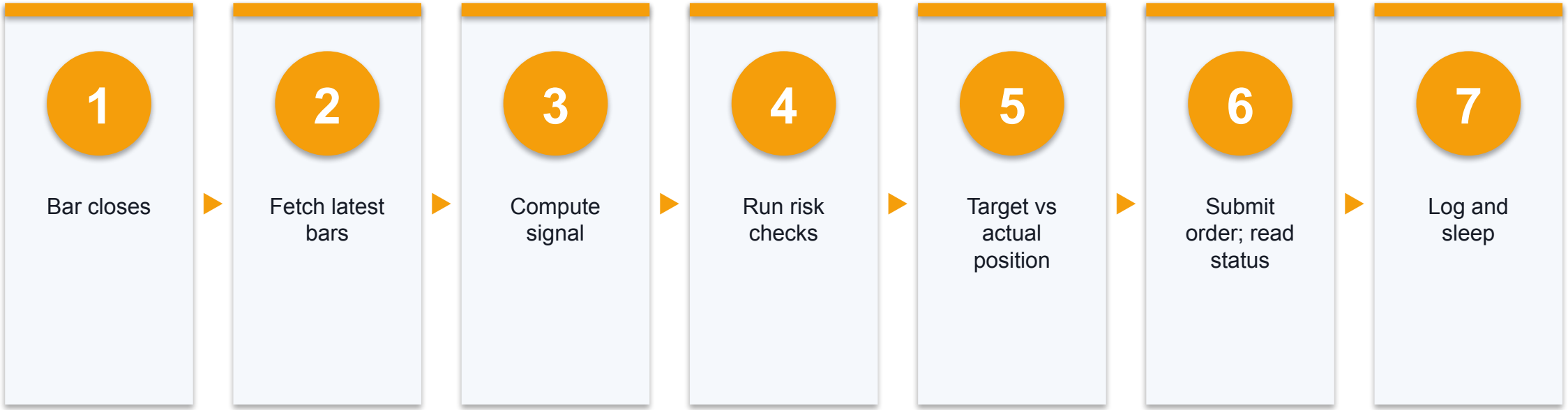


Configuration — symbols, parameters, limits, keys — lives outside the code.

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The Automated Paper-Trading Bot Loop

One cycle per closed bar.

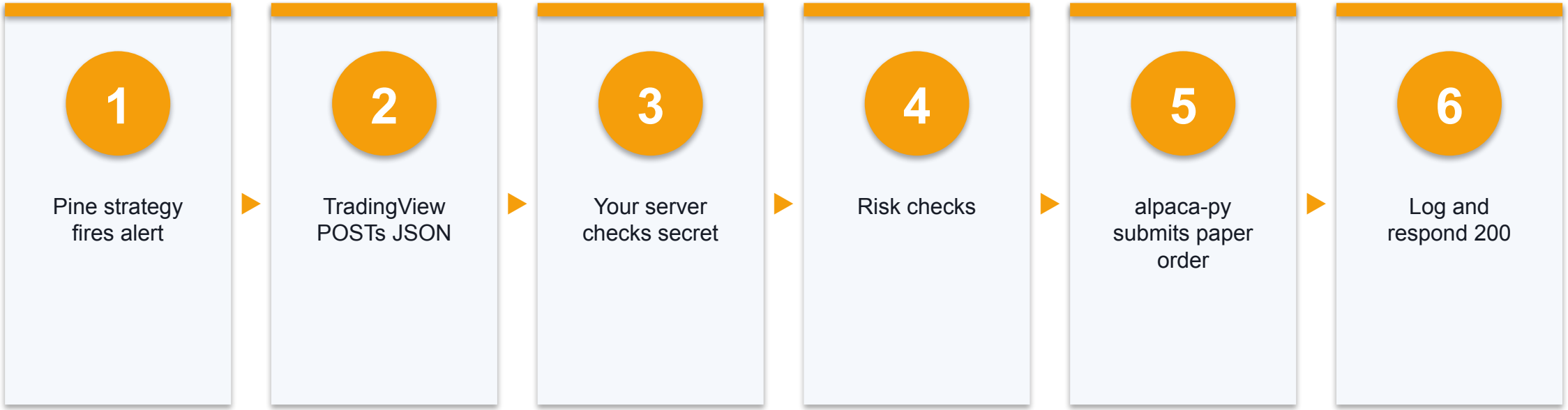


Every cycle writes to the trade log — the evidence for performance and audit.

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TradingView Alert to Alpaca Order

Webhook-driven execution.



Webhook alerts need a paid TradingView plan and a public URL (e.g. a tunnel).

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Diagnosing an Unintended Outcome

1. Reproduce the result
2. Isolate the first divergent step
3. Data · signal · order · fill?
4. Compare with a known-good run
5. Fix and re-test

Read the first failure, not the last — everything after it is a consequence.

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Lunch Break

1 hour · see you at 13:30

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A7 A8 K7

Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even

LAB 15

Run the same strategy under four cost models and find its break-even cost. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Zero vs Alpaca vs high-cost
2. Simulator: Break-even and execution timing
3. Simulator: Stops and sizing
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

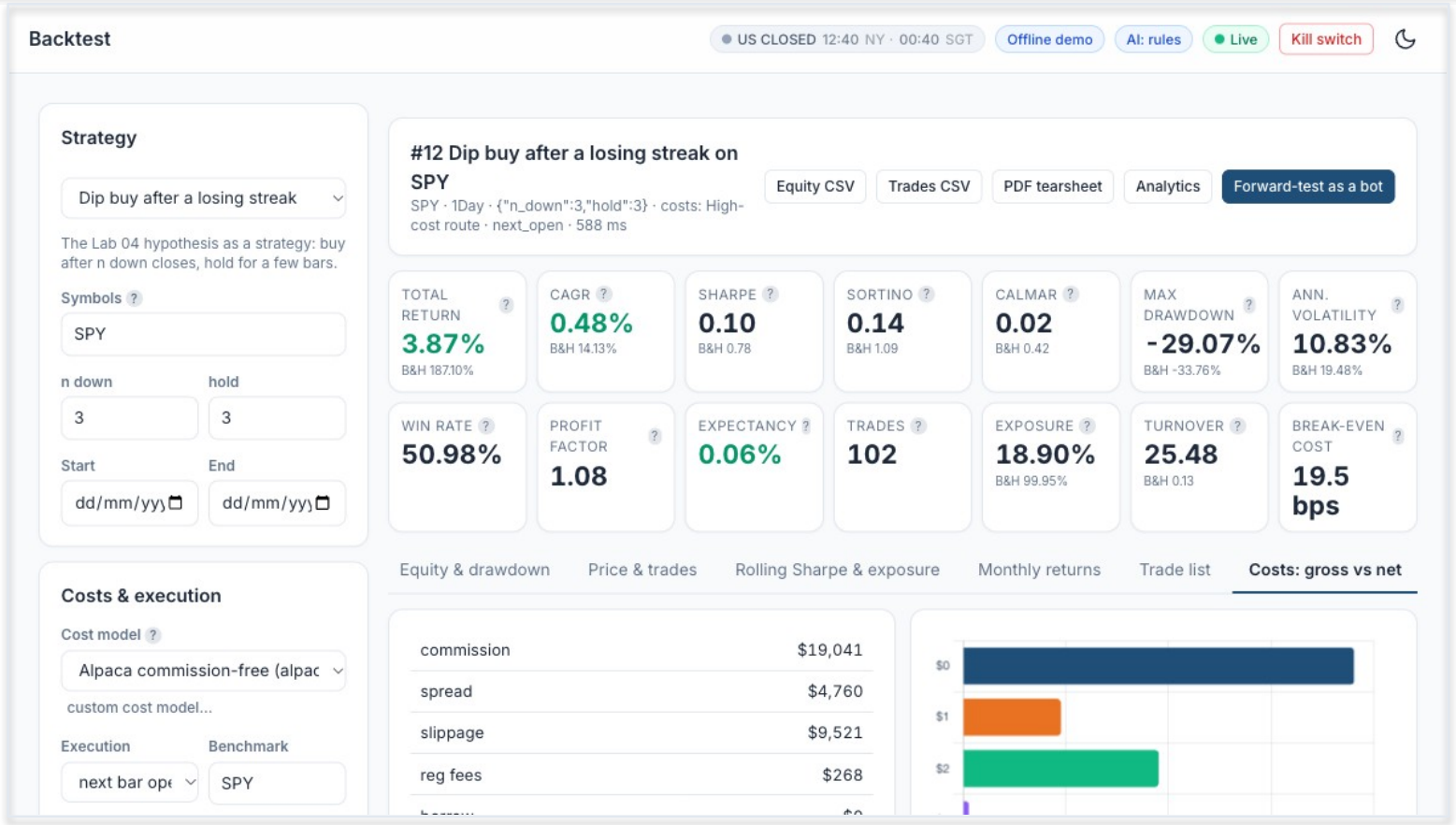
A backtest engine with three cost models, execution timing and break-even cost.

Duration
25 min

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LAB 15 · SIMULATOR

Lab 15 — In the Trading Simulator



Course Labs > Lab 15 at http://localhost:8000 - Backtest SPY dip-buy, high-cost route. Offline demo data; connect Alpaca paper keys for live paper trading.

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A7 A8 K7

Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading

LAB 16

Create a bot, run it on a schedule, watch orders, positions and P&L live, and halt it with the kill switch. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Create the bot
2. Simulator: Start it and watch
3. Simulator: Kill switch
4. Simulator: Backtest vs forward
5. Code: rebuild the stage with your AI assistant

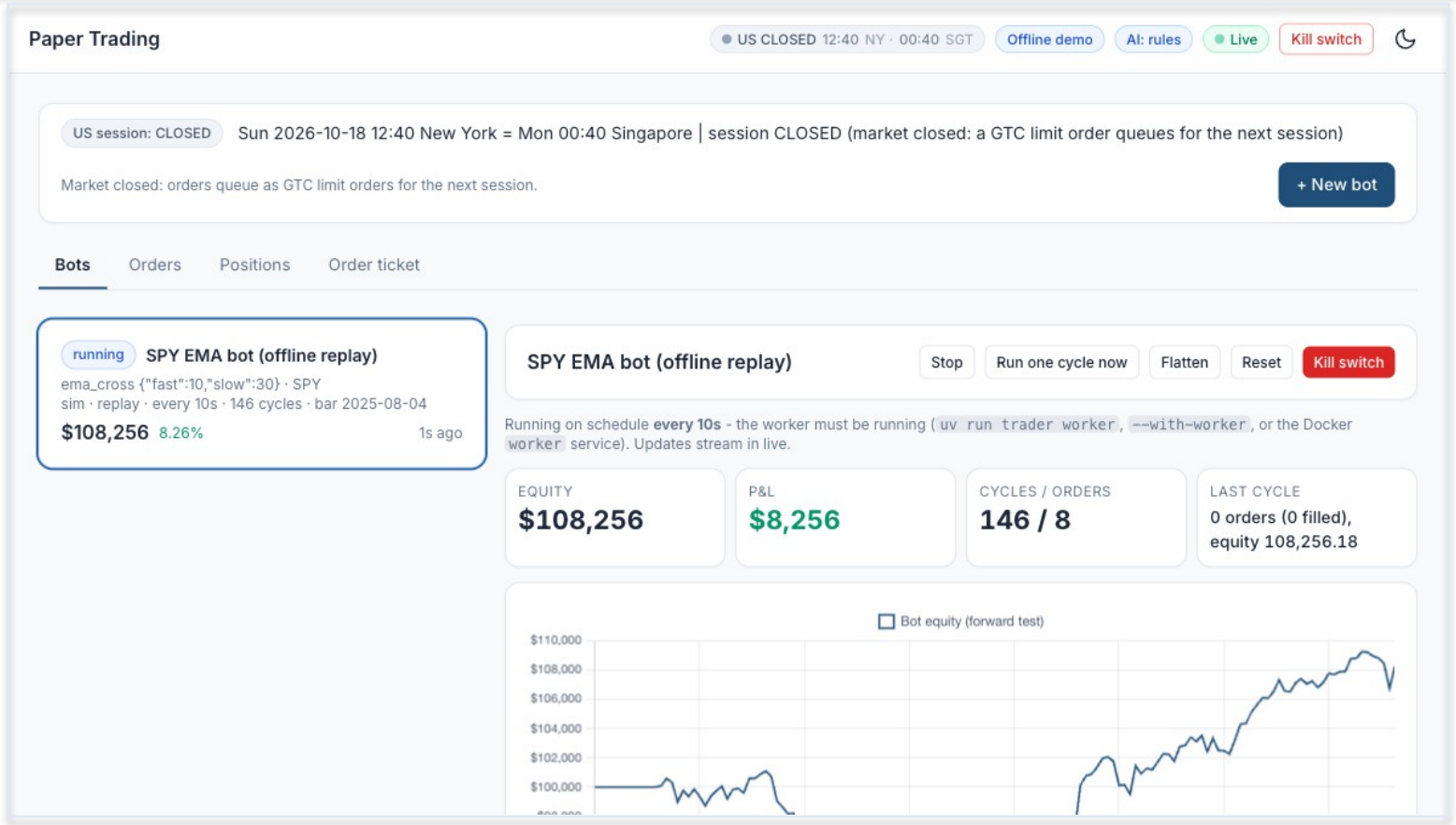
YOU'LL PRODUCE

A scheduled, risk-checked bot trading the paper account live, with a trade log and kill switch.

Duration
25 min

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Lab 16 — In the Trading Simulator



Course Labs > Lab 16 at http://localhost:8000 - Paper Trading > New bot (EMA cross). Offline demo data; connect Alpaca paper keys for live paper trading.

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A7 K5 K7

Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution

LAB 17

Send TradingView alerts to the app's webhook and see them pass validation, signal confirmation and risk into paper orders. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Copy the alert template
2. Simulator: Run the self-test
3. Simulator: Real alerts (paid TradingView plan)
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

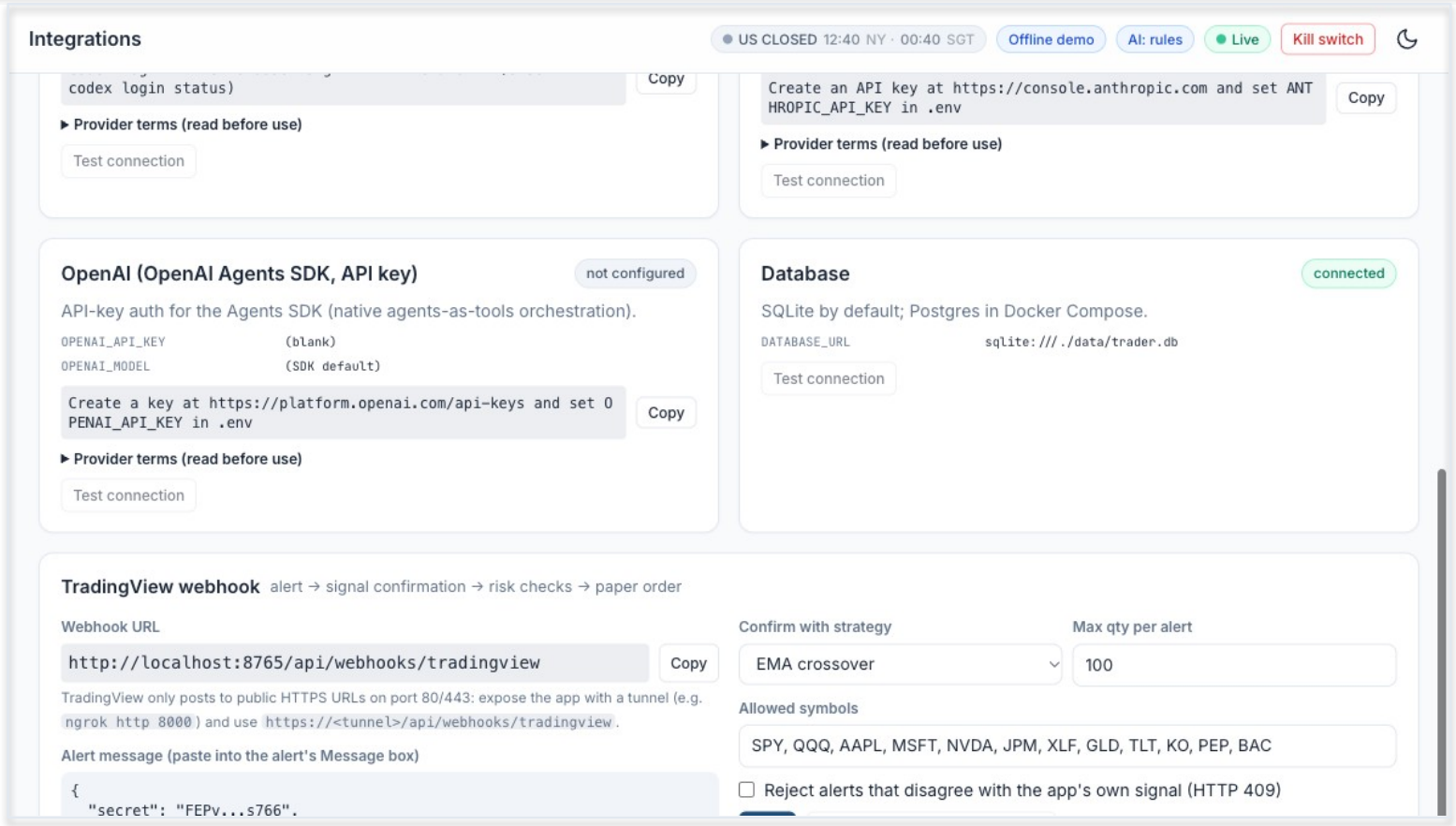
Event-driven automation: TradingView alert to webhook to signal, risk and paper execution.

Duration
20 min

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LAB 17 · SIMULATOR

Lab 17 — In the Trading Simulator



Course Labs > Lab 17 at http://localhost:8000 - Integrations > TradingView webhook. Offline demo data; connect Alpaca paper keys for live paper trading.

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A9 K8

Diagnostics Stage: Automated Bias Checks before Go-Live

LAB 18

Run the automated diagnostics on a deliberately biased strategy, fix it, and re-run. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Diagnose the biased exhibit
2. Simulator: Fix it in the Strategy Builder
3. Simulator: Re-run diagnostics
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

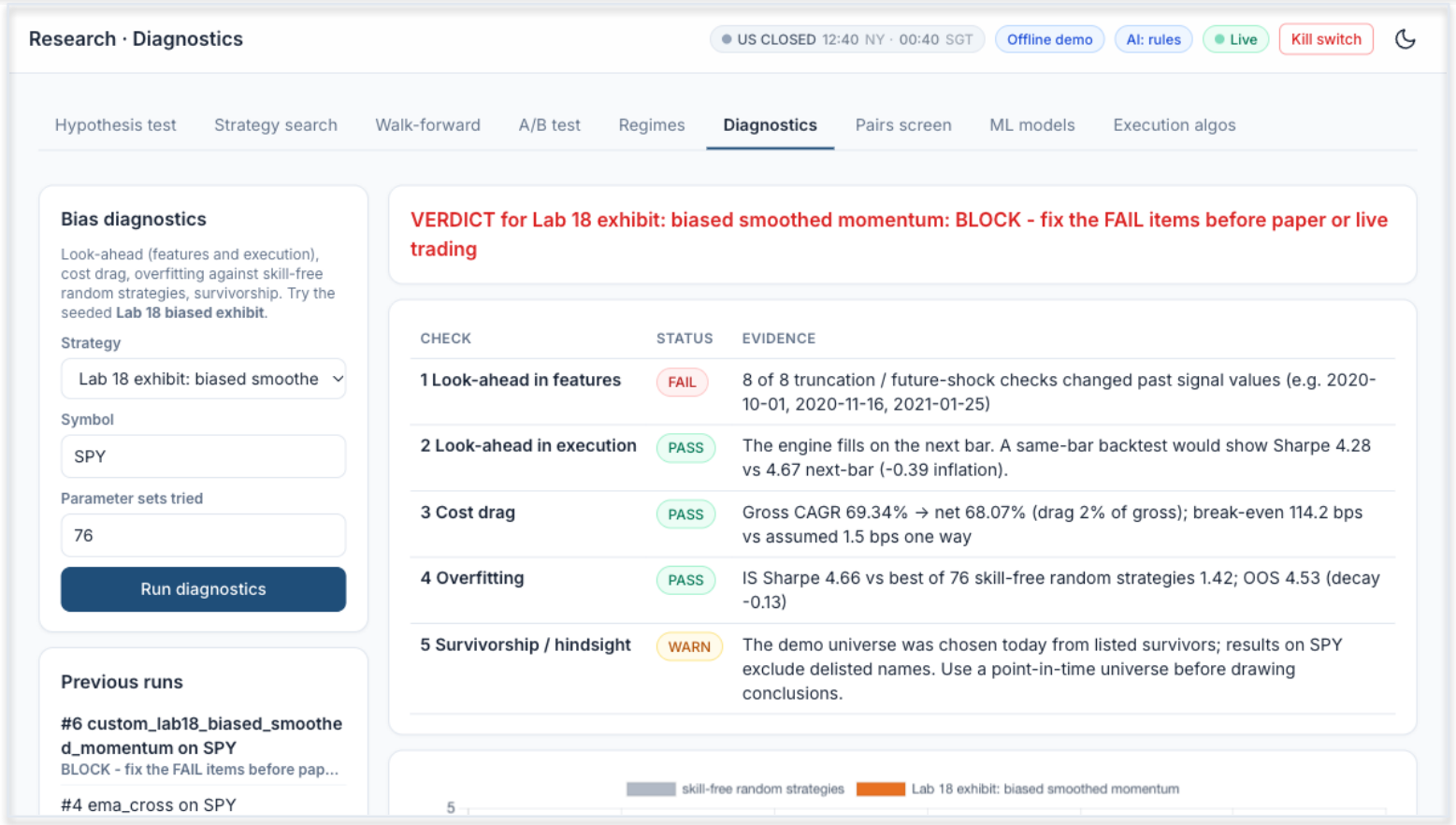
Automated bias diagnostics (look-ahead, costs, overfitting, survivorship) before go-live.

Duration
20 min

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LAB 18 · SIMULATOR

Lab 18 — In the Trading Simulator



Course Labs > Lab 18 at http://localhost:8000 - Research > Diagnostics (Lab 18 exhibit). Offline demo data; connect Alpaca paper keys for live paper trading.

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Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

LAB 19

Produce a one-page tearsheet and explain the gap between backtest and forward test. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Export the tearsheet
2. Simulator: Backtest vs forward
3. Simulator: Draw insights
4. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

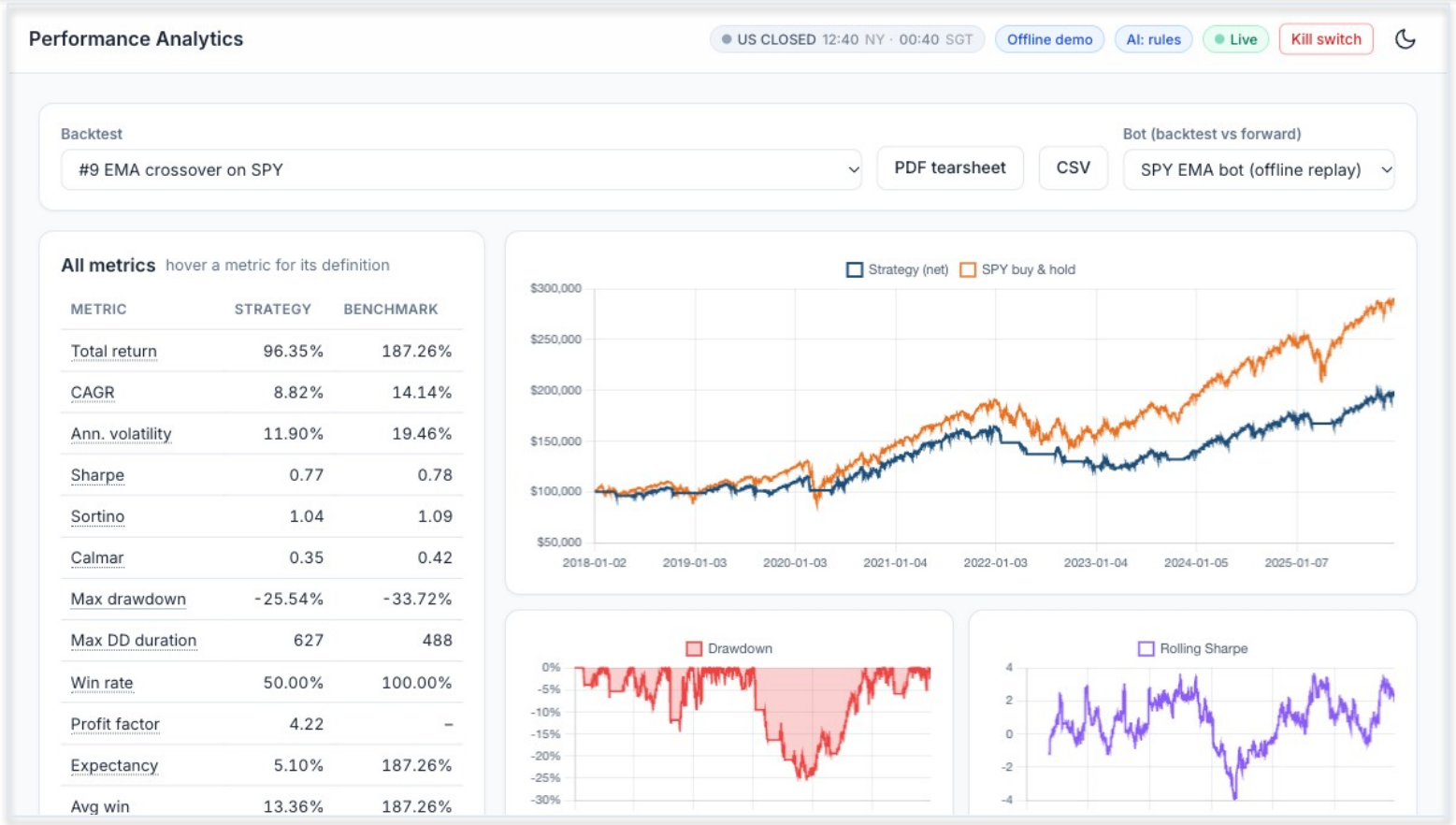
An automated tearsheet and backtest-versus-forward-test report with monitoring checks.

Duration
25 min

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LAB 19 · SIMULATOR

Lab 19 — In the Trading Simulator



Course Labs > Lab 19 at http://localhost:8000 - Analytics > PDF tearsheet. Offline demo data; connect Alpaca paper keys for live paper trading.

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A7 A10 K1

Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval

LAB 20

Run the whole pipeline end to end, get a human-approved proposal from the multi-agent desk, switch on the always-on agent, and write, backtest and forward-test your own Python strategy. Then build the same stage in code with your AI assistant.

WORKFLOW

1. Simulator: Run the multi-agent desk
2. Simulator: Human approval
3. Simulator: Always-on agent
4. Simulator: Write your own strategy
5. Simulator: Backtest, diagnose and forward-test it
6. Code: rebuild the stage with your AI assistant

YOU'LL PRODUCE

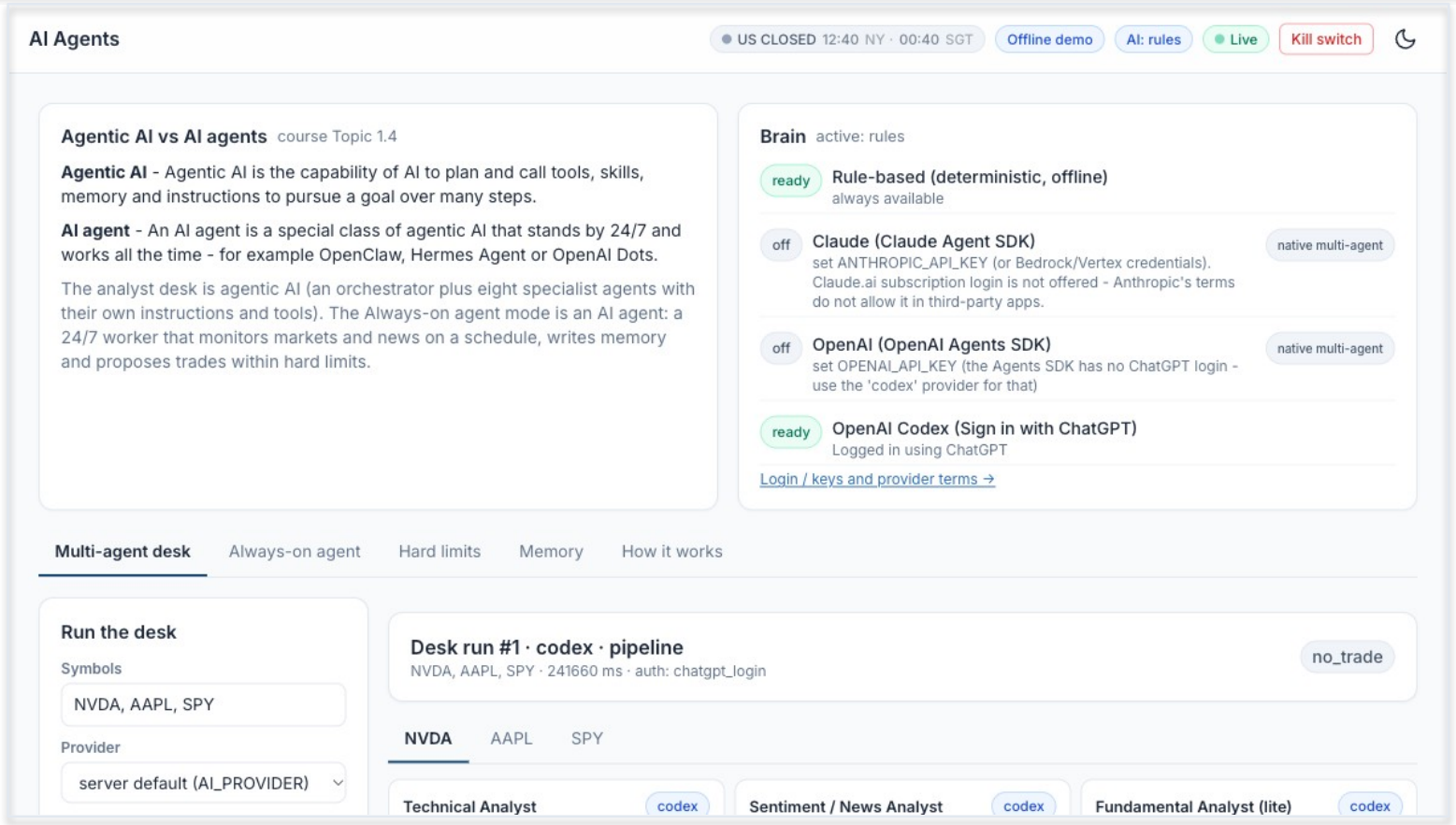
One command running the whole pipeline end to end, with a human-approved agentic layer.

Duration
20 min

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LAB 20 · SIMULATOR

Lab 20 — In the Trading Simulator



Course Labs > Lab 20 at http://localhost:8000 - AI Agents > desk. Offline demo data; connect Alpaca paper keys for live paper trading.

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A7 A8 A9 A10 K7 K8

Topic 4 Recap

1. **A7**
   Perform coding and configuration of software agents or programs based on selected models or algorithms
2. **A8**
   Conduct tests on the actions taken and outcomes to assess effectiveness of the models
3. **A9**
   Diagnose unintended outcomes produced by statistical models
4. **A10**
   Draw relevant trends and insights from data analysis to support decisions and report accordingly
5. **K7**
   Coding languages for programming of algorithms and signals
6. **K8**
   Potential reasons for unintended outcomes

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SUMMARY

What You Achieved

1. **Framed testable hypotheses**
   Instrument, signal, holding period, effect, benchmark.

2. **Coded indicators and strategies**
   Pine Script on TradingView; Python on Alpaca data.

3. **Modelled and A/B-tested**
   ML classifiers, regimes, Welch's t-test and bootstrap.

4. **Backtested net of costs**
   Commission, spread, slippage and regulatory fees.

5. **Forward-tested automatically**
   A paper-trading bot with risk checks and a trade log.

✓ **Diagnosed and reported**
Found the bias, fixed it, and wrote the recommendation.

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COURSE PORTAL

Courseware & Assessment on the LMS

Everything for the open-book assessment — slides, Learner Guide and submissions — lives on the LMS.

T

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LMS for IBF-STS courses

Sign in / OTP

lms-tms.tertiaryinfotech.com

1 **Sign in**
Log in with your registered email (OTP or password).

2 **Download the courseware**
Trainer slides, Learner Guide and lab artefacts.

3 **Mark attendance**
Sign in for each session — IBF-STS funding needs 75% attendance.

4 **Submit the assessment**
Upload your WA answers and PP evidence here.

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ASSESSMENT

Assessment — Reminder

Written Assessment

45 minutes · open book · 9 short-answer questions covering K1 - K9.

Practical Performance

75 minutes · open book · 4 tasks in Python with Alpaca paper trading, A1 - A10.

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ON ASSESSMENT DAY

Assessment Flow

1. Sign the assessment attendance record
2. Assessor briefing (not assessed)
3. Sit the WA (45 min) then the PP (75 min) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

Sign the assessment attendance record BEFORE you begin. Submit on the LMS.

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CLOSING

Before You Leave

1

Attendance

Make sure every session is signed — 75% is required for IBF-STS funding.

2

Course feedback

Complete the feedback form on the LMS.

3

Keep practising on paper

Your Alpaca paper account and lab code are yours to keep.

!

Not investment advice

Scenarios are fictitious; trade real money only with proper advice.

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Thank You

Machine Learning 101 for Financial Trading · FSE-DAT-3019-1.1

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